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## Enhanced air quality prediction using adaptive residual Bi-LSTM with pyramid dilation and optimal weighted feature selection

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In most industrial and urban regions, monitoring and safeguarding the air's purity is considered one of the most crucial tasks for government agencies. In numerous industrial and urban locations, preserving and tracking the condition of the air has become the primary concern. However, implementing comprehensive air quality monitoring systems often requires significant financial investment. The performance of current air quality monitoring sensors declines over time, which leads to inaccurate measurements of air pollution levels. To address this challenge, it is essential to develop and implement strategies aimed at maintaining sensor accuracy and effectively resolving environmental issues related to air quality. To facilitate an effective air quality prediction and assessment, a deep learning network is proposed. At first, the data for predicting air quality are collected from the relevant data sources. The proposed model introduces a novel methodology for weighted feature selection utilizing an Improved Gannet Optimization Algorithm (IGOA) aimed at enhancing the performance of data classification. After extracting the weighted features, classification is carried out using an Adaptive Residual Bi-LSTM network combined with Pyramid Dilation (ARBi-LSTM-PD), which significantly increase the model's potential to identify complex patterns within the data. The efficacy of the implemented model is enhanced by optimizing the parameters from RBi-LSTM using the IGOA strategy. This approach tackles the difficulties associated with feature selection and classification, leading to distinct advancements in the quality of the classification results. The robustness of the model is examined and analyzed using different measures. The accuracy and precision rate of the proposed model are 95.175% and 87.2%, which is better than traditional air quality prediction models. Thus, the simulation results demonstrate that it obtains the desired results for predicting and assessing the quality of air.

**Keywords** Air quality prediction, Health assessment, Optimal weighted feature selection, Real-time air data, Adaptive residual Bi-LSTM network with pyramid dilation, Improved gannet optimization algorithm

A system that uses real-time data from numerous air quality sensors to forecast the air quality in a specific location is often referred to as a real-time air quality prediction system<sup>1</sup>. This system captures information that includes Sulphur dioxide (SO<sub>2</sub>), and Particulate Matter (PM). These types of information help to predict the upcoming levels of air quality based on previous information and other environmental factors<sup>2</sup>. In most cases, the real-time air pollution forecasting system is implemented as an online program that provides a simple interface view for predicting indoor air quality levels<sup>3</sup>. If the air conditions go beyond the permitted limits, this network notifies users via electronic warnings or notifications. In general, a real-time air quality forecasting model is useful for monitoring and analyzing the quality of air in a particular location<sup>4</sup>. This technology can assist people in taking precautions to avoid air pollution and reduce the health hazards associated with it by providing reliable and precise air quality estimates obtained in real-time<sup>5</sup>.

Protecting the reliability of the sensor data is one of the main difficulties in a real-time air pollution forecasting system<sup>6</sup>. Climate patterns and alterations in the environment influence the sensors used to collect

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information regarding air quality<sup>7</sup>. The effectiveness of the system is maintained by ensuring the data's accuracy and dependability<sup>8</sup>. To ensure the systems' accuracy and reliability, significant quantities of historical data are needed for training the network<sup>9</sup>. To make accurate predictions in real time, these models must be capable of adjusting the shifting environmental variables<sup>10</sup>. The number of sensors required to survey an extensive area might be enormous, and processing the information consumes more computational resources. This system must be built to expand the number of sensors and data channels to handle enormous amounts of data<sup>11</sup>. Increasing public knowledge and encouraging the implementation of the system is considered another difficulty. The program must be simple to use and open to the entire audience<sup>12</sup>. The program should be user-friendly and accessible to everyone<sup>12</sup>. Implementing this educational initiative is crucial for raising public awareness about poor air quality.

To evaluate the huge amounts of data gathered from sensors, effective deep-learning approaches are frequently employed in real-time pollution forecasting systems<sup>13</sup>. A deep learning algorithm known as Artificial Neural Networks (ANNs) is capable of predicting complex relationships within data. ANNs are utilized to analyze information from various sources, including weather data, satellite imagery, and ground-based sensors<sup>14</sup>. Support Vector Machines (SVMs), which are a supervised learning method, can be used for both regression and classification tasks. SVMs are applied in real-time air pollution forecasting systems to analyze data from multiple devices, producing predictions based on historical data<sup>15</sup>. The accuracy of forecasts is impacted by the quality of the real-time data utilized. Traditional models often struggle to adapt dynamically to new data patterns or changing environmental conditions<sup>16</sup>. Combining optimal weighted feature selection with deep methods introduces computational challenges, particularly when handling large datasets. Without the incorporation of advanced deep learning techniques, it becomes difficult to recognize the efficacy of the proposed framework by addressing air quality issues. It is vital to implement robust validation processes to assess model effectiveness in real-world scenarios<sup>17</sup>. Hence, the proposed model developed optimal feature selection techniques to improve model outcomes by isolating the most significant factors influencing air quality. The creation of a reliable air quality forecasting model that utilizes optimal weighted feature selection facilitates greater accuracy in predictions. With precise air quality forecasts, decision-makers and environmental organizations execute timely and impactful measures to lower pollution levels and improve air quality, which is essential for promoting environmental sustainability.

The contributions of this work:

- A robust air quality prediction model is designed to improve the precision of air quality forecasts. This model employs optimal weighted feature selection to identify and control the most pertinent characteristics from the dataset, thereby enhancing predictive performance in contrast to models that indiscriminately utilize all available features. Air quality data are complex and multidimensional in nature, integrating optimal weighted feature selection process updates the data by filtering out less significant features, resulting in a more efficient and interpretable model.
- An IGOA strategy is designed for selecting the relevant features from the raw data. This approach utilizes evolutionary techniques to identify the most critical features from the data. By fine-tuning the choice of features along with their assigned weights, the IGOA boosts the precision and performance of prediction models. The selected features and their weights subsequently are applied in future predictive models to ensure that only the most important features are considered for predictions, which ultimately enhances the proposed model's accuracy and interpretability.
- An ARBi-LSTM-PD model is developed for air quality prediction tasks. It forecasts air quality by proficiently utilizing historical data, emphasizing the importance of long-term dependencies and contextual information through bidirectional processing. The integration of residual connections improves training performance, while the adaptive component enables the model to adjust to real-world changes. Additionally, pyramid dilation boosts its ability to identify features across various scales. Together, these elements contribute to more reliable and precise predictions of air quality. By incorporating RBi-LSTM and pyramid dilation, the proposed model achieves a better balance between complexity and performance, reducing the overfitting issues while capturing essential patterns in the data.
- The unique contribution of this proposed framework lies in the utilization of IGOA for the processes of weighted feature selection and subsequent classification through an ARBi-LSTM-PD. This optimization technique improves predictive performance by selecting and assigning proper weights to the most pertinent features. This innovative methodology ensures that the proposed framework not only identifies key air quality metrics but also evaluates health risks linked to different levels of air pollution.

### Novelty of proposed model

The model presents an innovative method for feature selection by utilizing the IGOA focused on weighted feature selection. This strategy enables to select most pertinent features while allocating optimal weights, thereby improving the model's capability to identify critical data attributes that greatly impact predictions. This integration enhances the model's ability to hold temporal dependencies and complex patterns within air quality data, resulting in better classification results than traditional frameworks.

### Motivation of this proposed model

The unique motivation behind this research is to improve the precision and dependability of air quality forecasting through the development of a novel framework that merges optimal weighted feature selection with sophisticated deep learning approaches. The main aim is to tackle the intrinsic challenges and multifaceted aspects of air quality data by utilizing a structured method to eliminate less relevant features, thus strengthen the model's clarity and effectiveness.

The sections involved in describing the proposed air quality prediction model are specified as follows. Subdivision II specifies the review of the traditional quality prediction system along with its features and limitations. Subdivision III demonstrates the requirement for air condition assessment and health assessment in smart cities. In Subdivision IV, the benefits of a heuristic-assisted algorithm for the selection of relevant features are briefly discussed. Implemented ARBi-LSTM-PD-based air quality prediction and health assessment are discussed in Subdivision V. Subdivision VI presents an outcome and the discussion of the IGOA-RBi-LSTM-PD-based air quality prediction. The quick summary and research limitations are provided in Subdivision VII.

## Literature survey

### Related works

In 2024, Natarajan et al.<sup>18</sup> developed an innovative regression model for forecasting the AQI by integrating the optimization algorithm with a Decision Tree regression model, which enhances prediction accuracy. The approach included the extraction of key features from the dataset utilizing the optimization algorithm, enabling the identification of the most significant factors influencing AQI predictions. A thorough experimental analysis was performed using well-documented air pollution datasets.

In 2023, Van, et al.<sup>19</sup> have developed a machine learning-based technique for determining the quality of air. Here, three machine learning techniques were employed to forecast the AQI. The database acquired from the Indian location was adopted to identify the efficiency of the system. The simulation results showed that the AQI prediction could be effectively done by the XGBoost system. Moreover, the XGBoost system consumed minimum cost for predicting the AQI.

In 2022, Tella, et al.<sup>20</sup> have suggested spatial air quality maps for the prediction of the quality of air. It was formed by merging the Geographical Information System (GIS) and machine learning techniques. The quality could be easily estimated by the spatial air quality maps. This system was taken in two countries, namely Selangor and Southeast Asian countries for maintaining the quality of air in the future environment.

In 2025, Dongre et al.<sup>21</sup> introduced an integrated framework for predicting air quality that combined multiple outlier detection methods with machine learning models, thereby improving prediction accuracy and robustness. It examined various outlier detection techniques to assess their effects on model performance. It highlighted the significance of robust outlier management in the training of models and the improvement of their accuracy.

In 2022, Baqer et al.<sup>22</sup> developed a machine learning technique for assessing the air pollutants of the environment. The proposed system has two stages. The IAQ dataset was utilized in the first step of the developed method and the weight values were assigned for all pollutants in the environment while executing the second step. The pollutants were classified using the Unified Process for Labeling Pollutants Dataset (UPLPD), and it was attained from the IT2TR-FWZIC. Lastly, five performance indices were considered to evaluate the invented model.

In 2022, Asif et al.<sup>23</sup> offered an Adversarial Sequence Tagging system, and it was used for the determination of porosity and porosity-excessive penetration in the environment. Here, the logistic regression algorithm and the sequence tagging system were accompanied by the developed model to attain efficient predicted outcomes. Moreover, the exactness of the proposed model was improved when utilizing sequence tagging with the suggested model.

In 2023, Nilesh et al.<sup>24</sup> proposed a machine learning technique for an air pollution assessment system. The input given to the proposed model was maintained by several preprocessing techniques. The data required for this prediction was collected from the Central Pollution Control Board of India. In this work, the SVM with RBF kernel was utilized to perform the prediction procedure for previously collected data. The simulation outcome showed that the efficiency of the offered system was enhanced than the other classifier.

In 2025, Qian et al.<sup>25</sup> introduced an innovative hybrid model for predicting the AQI by combining TimesNet and Crossformer, which improved predictive performance over conventional models. Additionally, the research conducted a comprehensive comparison of various weighting schemes and benchmarks against established models, revealing notable gains in accuracy metrics across multiple datasets from various cities.

The research articles from<sup>26–30</sup> demonstrate the adaptability of machine learning across diverse domains. AdaBoost has been effectively used for feature selection and classification, often in relation to SVM<sup>26</sup>, while CNN+LSTM hybrid models improve air quality prediction by capturing spatial-temporal patterns<sup>27</sup>. In healthcare, re-sampling with AdaBoost enhances classification of imbalanced ECG beats<sup>28</sup>, and deep CNNs with Bayesian learning enable accurate plant leaf disease detection for agriculture<sup>29</sup>. Furthermore, novel ML frameworks support quality prediction and control in wire arc additive manufacturing<sup>30</sup>. Overall, these works underline the impact of boosting, hybrid deep learning, and advanced frameworks in improving accuracy, efficiency, and decision-making.

Recent studies have applied machine learning and optimization techniques across diverse domains. Zaki<sup>31</sup> employed KNN to predict air quality in Delhi, while<sup>32</sup> reviewed ML models for urban air quality forecasting, outlining their strengths and gaps. In intelligent transportation, Alhussan et al.<sup>33</sup> integrated adaptive mutation optimization with transfer learning for pothole classification in self-driving cars. Salamai et al.<sup>34</sup> enhanced supply chain 4.0 risk management through a dynamic voting classifier. Furthermore, El-Sayed et al.<sup>35</sup> introduced the Greylag Goose Optimization algorithm, demonstrating improved convergence and solution quality over conventional metaheuristics.

### Discussion based on previous works

These existing models highlight a significant advancement in predicting air quality by incorporating advanced methodologies, effective algorithms, and thorough attention to data attributes, such as outliers and spatial considerations. As the dependence on these predictive models for monitoring environmental conditions increases, subsequent research should aim to enhance these approaches further and investigate new strategies to

tackle the intricacies of air quality management. The combination of deep learning networks and data handling techniques will surely influence the evolution of precise air quality forecasting in the future.

### Research gaps in the existing research

Several existing models do not undergo thorough validation using a diverse set of evaluation metrics. Creating a benchmark for comprehensive validation might improve the reliability of results and enable more effective comparisons between models. Additionally, many existing models concentrate on particular geographic areas, which can restrict their general applicability. Furthermore, there is a notable absence of real-time data incorporation in current predictive models, which is essential for prompt AQI evaluations. Tackling these difficulties could result in considerable progress in the forecasting abilities and practical uses of AQI prediction models, thereby improving strategies for managing air quality.

### Problem statement

Air quality prediction involves forecasting the levels of airborne contaminants with the support of different modelling approaches and data analytical techniques. This forecasting is essential for analyzing the patterns of air quality and their effects on public health. A range of statistical and machine learning methods are employed to make these predictions regarding air quality. Yet, in traditional models, the air quality datasets frequently display nonlinearity due to a variety of influencing factors, such as climatic conditions, traffic dynamics, and emissions from industries. Many conventional statistical models find it challenging to accurately represent these intricate interactions. Additionally, the presence of inconsistent or incomplete data adversely affects the reliability of predictions. Issues such as absent values, inaccuracies in data gathering, and differences in monitoring methods across diverse areas complicate the analytical process.

Some of the common challenges that exist in traditional models, as well as how the proposed model addresses these issues, are outlined below.

- In traditional models, finding the most pertinent predictors within large datasets poses a significant challenge. Irrelevant features reduce the effectiveness of the model, whereas the absence of crucial variables results in unreliable predictions. This challenge is tackled through the implementation of weighted feature selection. This method enhances the selection process by focusing on the most valuable features while reducing the influence of irrelevant features. By utilizing an optimization strategy, the proposed model successfully eliminates less significant data, resulting in improved accuracy and predictive ability.
- Existing models are particularly susceptible to overfitting. This occurs when they excel with training data but struggle to apply their knowledge to unseen data, resulting in low predictive accuracy. Air quality fluctuates quickly due to varying environmental conditions. However, many existing techniques have difficulty in capturing these changes across various time scales. The proposed model addresses this issue by assimilating information efficiently from the input features while ensuring robust performance in a more complex network. The suggested attention mechanism enables the proposed model to recognize variations in air quality across different temporal scales, addressing the challenges faced by conventional methods that often fail to adequately capture such fluctuations.
- In existing models, the diversity of data sources makes the development of models more complex. Although machine learning techniques are highly effective, the real-time analysis of continuous data for instant prediction leads to technical difficulties. The suggested model tackles these issues by employing an improved deep-learning framework that effectively combines various data sources. This strategy improves the handling of diverse data types, simplifies the model development process, and enables quick predictions while maintaining a high level of accuracy.
- Thus, proposed model provides more dependable predictions, as critical variables are preserved and highlighted throughout the modeling process, thus addressing the shortcomings of conventional approaches.

The features and challenges of existing air quality prediction models are showcased in Table 1.

## Detailed description of developed air quality and health assessment framework in smart cities using deep learning

### Requirement of air quality prediction and health assessment in smart cities

In recent years, most of the countries in the world have been facing pollution issues due to climate and environmental alterations. Because of air pollution, the people present in the world are affected by severe health issues, which cause a high death rate. Sickness, such as respiratory issues and lung disease, are caused because of air pollution in the surrounding environment. People who are affected with chronic asthma disease are severely affected by the ozone and nitrogen oxide of the environment. The purity of the air is determined by AQI. So, it is important to monitor the AQI in smart cities to manage the quality of air. The lifestyle of the people is altered by adopting the smart city concept. Here, the quality of life is improved by employing sensor devices. These sensor devices can collect data from various sources and it can result in the development of weather alteration and pollution to the environment. The information and communication services in the smart city are the main reason for the provision of better communication and transportation facilities. In smart cities, the total amount of population is raised due to the above communication and transportation facilities. Transportation devices are enormously raised due to the increase in population. The vehicle used by the people present in smart cities emits lots of carbon dioxide into the environment, and therefore, the quality of air is greatly reduced, which can result in affecting the health condition of people. As the population in smart cities grows, the need for transportation similarly rises. A rise in the number of residents leads to more cars on the streets, resulting in greater emissions of harmful pollutants. These urban environments prioritize specific transportation methods that worsen air

| Author [citation]              | Methodology              | Features                                                                                                                                                                                                                                         | Challenges                                                                                                                                                                                                              | Performance metrics |
|--------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Natarajan et al. <sup>18</sup> | Regression model         | • It can analyze vast quantities of data much faster and more accurately than humans, making it easier to extract insights and patterns from large datasets.                                                                                     | • Overfitting occurs when a machine learning model is too intricate and fits directly to the training data, leading to poor performance when applied to new data.<br>• It requires significant computational resources. | • Accuracy is 88.9% |
| Van, et al. <sup>19</sup>      | XGBoost                  | • It is optimized for speed and efficiency, making it possible to train models on large datasets with high speed and low memory consumption. It can handle missing values in datasets, which is a common problem in many machine-learning tasks. | • It has many hyperparameters that need to be tuned for optimal performance, which can be time-consuming and require expertise.                                                                                         |                     |
| Tella, et al. <sup>20</sup>    | KNN                      | • It does not formulate any assumption about the fundamental data distribution. This makes it more flexible and robust than parametric models.                                                                                                   | • It may be slow when working with high-dimensional data since the distance calculations between data points become more computationally expensive.                                                                     | • Accuracy is 98%   |
| Dongre et al. <sup>21</sup>    | IQR                      | • It can handle missing data effectively by using the available data to make accurate predictions. It gives assess of quality magnitude, which can be used to make out the large amount of important features in the dataset.                    | • It can be slow to train, especially for large datasets with many features.                                                                                                                                            | • MAE is 11.9       |
| Baqer et al. <sup>22</sup>     | ML                       | • It has the potential to handle more amount of data quickly and accurately, which can save time and money and increase efficiency in various industries.                                                                                        | • Developing and implementing machine learning models is time-consuming, requiring specialized expertise and computing resources.                                                                                       | • Accuracy is 99.8% |
| Asif et al. <sup>23</sup>      | XGBoost                  | • It can handle missing values in datasets, which is a common problem in many machine-learning tasks.                                                                                                                                            | • It is prone to overfitting if the hyperparameters are not properly tuned or if the dataset is too small.                                                                                                              | • Accuracy is 91.1% |
| Nilesh et al. <sup>24</sup>    | SVM                      | • It is useful for regression problems, and it can handle both binary and multi-class problems.                                                                                                                                                  | • It can be complicated to construe, making it tough to recognize how the algorithm is making its predictions.                                                                                                          | • MAE is 18.6       |
| Qian et al. <sup>25</sup>      | TimesNet and Crossformer | • It is used to analyze the relationships between multiple independent variables and a dependent variable. This makes it a versatile tool for solving a wide range of problems.                                                                  | • It can be sensitive to outliers, which can skew the results and lead to inaccurate predictions.                                                                                                                       | • MAE is 22-35%     |

**Table 1.** Features and challenges of existing air quality prediction system.

quality, such as private vehicles. These factors underscore the complexity of maintaining air quality in smart cities as they navigate growth and transformation. Effective urban planning and smart technologies are essential to mitigating the negative impacts on air quality associated with population growth and transportation in smart urban environments. So, there is a need to expand air excellence prediction and the health assessment system for solving the above issues.

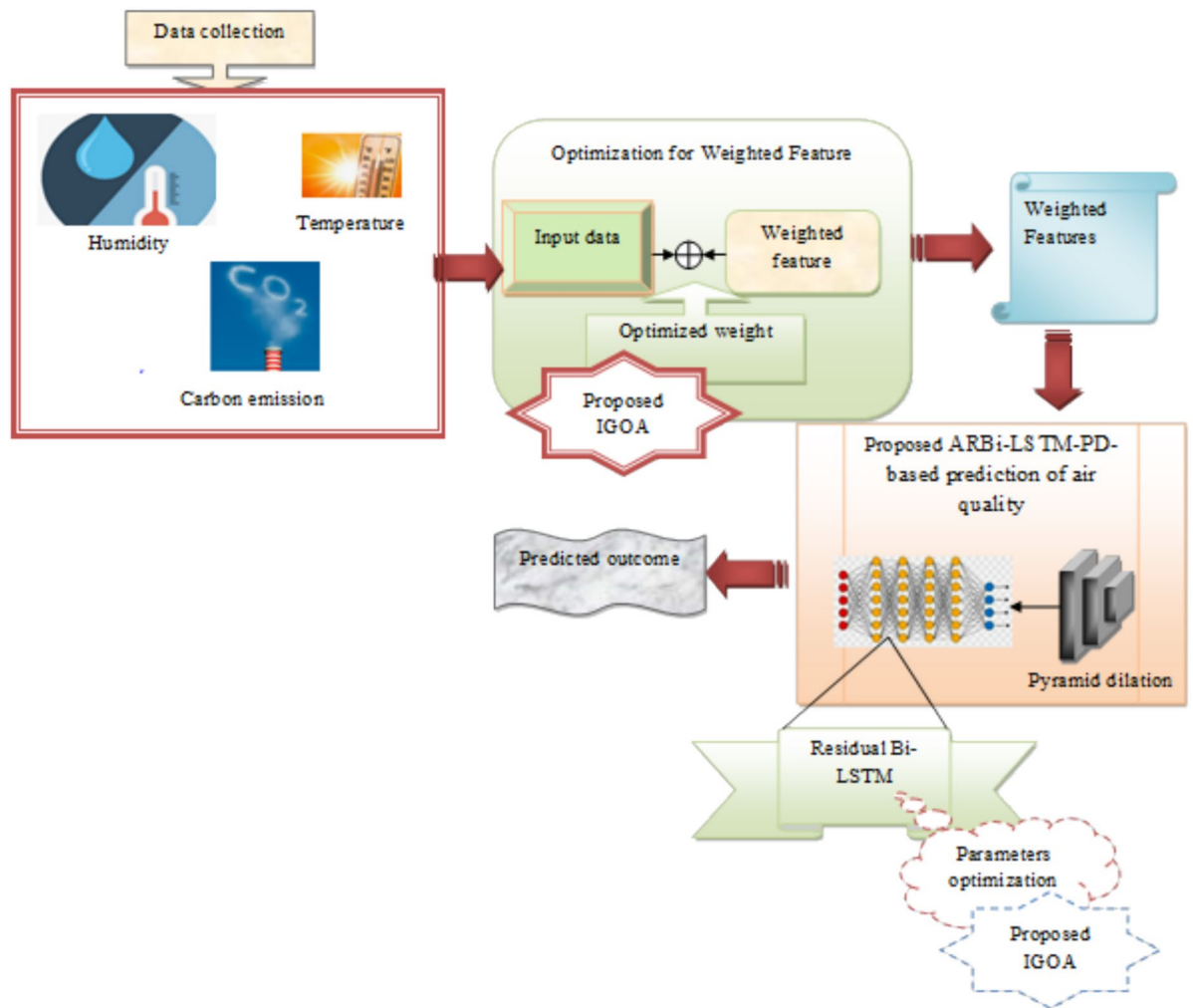
**Developed air quality prediction and health assessment scheme**

The air quality prediction and health assessment scheme is designed to enhance public health results, provide practical guidance for managing environmental issues, and foster a healthier living environment by utilizing advanced forecasting technologies for air quality prediction. By forecasting pollution levels, the proposed model assists in evaluating the potential health risks of different pollution scenarios, thereby educating the public on precautions to adopt during periods of rising pollution. The procedure begins with the collection of raw data, which generally encompasses numerous features that commonly consist of various types of details, including environmental parameters, temperature, humidity, and other pertinent factors. An optimization algorithm IGOA is employed to analyze and select the appropriate features from the input data. This selection occurs through a series of iterative steps where the optimization algorithm explores different feature combinations, evaluates their performance, and identifies the optimal set that provides the most significant features for the prediction task. After selecting the relevant features, weights are assigned to quantify their influence on the predicted results. By multiplying the selected features with these weights, the proposed model concentrates more on crucial predictors while minimizing the effects less impactful. Finally, the selected features are fed to ARBi-LSTM-PD for the prediction of AQI. This proposed model processes data in both directions forward and backward facilitating a more effective capture of temporal relationships. The vanishing gradient issue is solved by the inclusion of residual connections, thereby simplifying the training process and potentially enhancing overall performance. Pyramid Dilation is added, which enables the network to broaden its receptive field without increasing the number of parameters or computational load. This characteristic is particularly advantageous in time series forecasting tasks, such as predicting AQI. Once the weighted features are processed through the ARBi-LSTM-PD network, the model produces an output that classifies the AQI into specific categories. The efficacy of the proposed ARBi-LSTM-PD model is boosted by optimizing the parameters from RBi-LSTM using IGOA. This optimization process boosts the accuracy of AQI prediction. The predicted AQI value is utilized for recognizing the purity level of the air in the surrounding environment. Because of the air excellence forecast process, the health condition of humans is also assessed. The diverse methods are evaluated with the outcome of the developed model to find its efficacy. The diagrammatic illustration of the developed air quality prediction model is illustrated in Fig. 1.

**Collection of air quality dataset**

Dataset 1: The required data for the air quality assessment process is collected from the link of <https://github.com/adityarc19/aqi-india> access date: 2025-2-14. This dataset encompasses air quality information covering the years 2015 to 2020, facilitating the examination of changes over time and patterns in air pollution across various areas of India. AQI figures are classified into six distinct categories (such as Good, Satisfactory, Moderate, Poor, Very Poor, and Severe), supporting the evaluation of whether the air quality is suitable or presents potential health hazards.





**Fig. 1.** Diagrammatic illustration of the developed air quality assessment.

Dataset 2: The desired data are collected from the Real-time Air Quality Index from the link of <https://data.gov.in/resource/real-time-air-quality-index-various-locations> with access: 2025-2-14. Real-time AQI gathers data from air quality monitoring stations that constantly assess pollutant concentrations, ensuring the information accurately represents the present conditions. Numerous areas make use of real-time AQI data to issue notifications and warnings. When AQI values hit dangerous levels, alerts may be distributed to inform the public about the possible dangers linked to outdoor activities.

Dataset 3: The information is taken from the Comparison-of-ML-models-for-predicting-AQI having the link <https://github.com/Anindya-Das02/Comparison-of-ML-models-for-predicting-AQI> with access: 2023-7-14. Here, the regression systems are adopted for identifying the distinction of air. It has the real time air condition of several countries. Here, the health condition of the individual is categorized based on the AQI table.

The acquired data from the air quality prediction dataset is implied as  $AP_d$ , here,  $d = 1, 2..D$  and  $D$  is the total data gathered from the dataset. The collected data is partitioned into 3:1 ratio for performing training and testing. The AQI of the surrounding environment is determined based on the meteorological parameters such as wind speed, air pressure, dew point, ambient temperature and relative humidity. The level of air quality is determined based on the predicted values of the AQI and then the health condition of the peoples at the surrounding environment is predicted using this developed model.

## Heuristic assisted weighted feature selection for developed air quality prediction and health assessment model

### Proposed IGOA strategy

#### Novelty

In this proposed air quality forecasting model, the IGOA strategy is employed for optimally selecting the weighted features from the raw data and also for optimizing the variables from the Bi-LSTM network, which enhances the prediction accuracy. Air quality prediction is considered as a time series forecasting challenge, involving the collection of air quality measurements at designated time points. The main goal of this model is to forecast upcoming air quality conditions by utilizing past data. The IGOA introduces enhancements, notably

the adaptive modification of the random variable linked to the effectiveness of candidate solutions, which helps overcome the conventional challenges faced by GOA. By enhancing the speed of convergence and decreasing the chances of becoming trapped in local optima, IGOA facilitates more precise and dependable forecasts of air quality, which are critical for assessments related to public health.

GOA<sup>36</sup>

A set of candidate solutions (or gannets) is randomly selected within the defined search space. Each gannet is represented as a possible solution, with its location usually characterized by a vector of parameters. The position of each gannet is assessed to ascertain its fitness, determined by the objective function pertinent to the optimization challenge. Parameters such as the total number of gannets, the maximum number of iterations, and other factors related to the diving patterns should be established. Thus the position of the individual is calculated using Eq. (1).

$$V_{p,k} = Q_r \times (Qu_k - Ql_k) + Ql \quad p = 1, 2..M \tag{1}$$

Here, the upper and lower boundary of  $K^{th}$  individual is indicated as  $Qu_k$  and  $Ql_k$ , respectively. The term  $Q_r$  is the random variable that varies from  $[0 - 1]$  and  $M$  is the dimension size of the problem. Yet, traditional GOA faces difficulties related to slow convergence toward optimal solutions, especially in high-dimensional settings. This algorithm gets stuck in local optima due to its established search mechanisms. To tackle these issues, the IGOA modifies the random variable  $Q_r$  utilized during the optimization process. Rather than relying on a fixed random variable, IGOA vigorously adjusts this variable based on the best and worst fitness values, which enhances the effectiveness of the search process. These alterations enable the algorithm to adapt more flexibly to various optimization problems, thereby enhancing its effectiveness in tasks such as feature selection and parameter optimization in Bi-LSTM networks. The advancements in IGOA achieved faster convergence rates and diminished the chances of becoming trapped in local optima, ultimately resulting in more reliable solutions for air quality forecasting and health assessments. The random variable  $Q_r$  will be modified using Eq. (2).

$$Q_r = \frac{best}{(best/worst) \div (worst/best)} \tag{2}$$

Here, worst fitness, and best fitness values are depicted as *best*, and *worst*, correspondingly. The term  $Q_r$  elucidates the random number.

IGOA is derived from the foraging behavior of gannets, who dynamically adjust their diving angles and directions based on environmental feedback to maximize prey capture. This adaptive search behavior is modeled through a self-adjusting random variable that evolves based on both best and worst fitness values. Such feedback-driven stochastic control promotes global exploration in early stages and local exploitation in later stages of optimization. Compared to conventional methods like GA, PSO, and GWO, which use static control parameters or predefined mutation strategies, IGOA dynamically modifies its exploration behavior during the optimization process. This leads to superior convergence stability and robustness in avoiding local minima, particularly valuable in high-dimensional feature spaces such as those encountered in air quality modeling. Empirical validation in our convergence plots (Section "[Convergence analysis of proposed model](#)", Figure 6) demonstrates that IGOA achieves faster and smoother convergence than its non-adaptive counterparts like GOA, POA, and GMO. A comparative analysis of characteristics of IGOA with baseline models has been provided in Table 2.

A significant dynamic attribute of IGOA is its ability to adjust the random variable related to the performance of potential solutions adaptively. In contrast to conventional GOA, which employs a static random variable, IGOA modifies this variable in response to both the highest and lowest fitness values observed in the candidate solutions. This capability enables the algorithm to better address different optimization challenges, which is essential when dealing with high-dimensional datasets. Air quality forecasting is approached as a time series challenge that necessitates the gathering of air quality data at predetermined time intervals. The main aim of this model is to predict air quality levels by utilizing past information. The IGOA method provides notable enhancements, especially through the dynamic modification of the random variable linked to the performance of potential solutions. This flexibility addresses common issues encountered with traditional GOA, such as sluggish convergence and local optima.

| Optimizer                           | Adaptivity of parameters        | Exploration–exploitation balance       | Premature convergence handling | Feature selection capability    | Convergence rate  |
|-------------------------------------|---------------------------------|----------------------------------------|--------------------------------|---------------------------------|-------------------|
| GA (Genetic Algorithm)              | Static crossover/mutation rates | Moderate (global search-focused)       | Moderate (can stagnate early)  | High (but needs tuning)         | Moderate          |
| PSO (Particle Swarm Optimization)   | Static/random inertia weights   | Skewed toward exploitation             | Prone to local optima          | Moderate                        | Fast but unstable |
| GWO (Grey Wolf Optimizer)           | Rule-based leadership hierarchy | Balanced, but fixed strategy           | Moderate                       | High                            | Moderate          |
| GOA (Gannet Optimization Algorithm) | Static random variable          | Moderate (dependent on initialization) | Weak control over local traps  | High                            | Slow              |
| IGOA ( <i>Proposed</i> )            | Dynamic feedback-based control  | High, adaptive across iterations       | Strong (avoids local optima)   | Very High (optimized weighting) | Fast and stable   |

**Table 2.** Comparative characteristics of IGOA and other optimization algorithms.

The pseudocode of the IGOA algorithm is given in Algorithm 1.

|                                                                              |
|------------------------------------------------------------------------------|
| Initialize population count $cp$ and the iteration number $inm$              |
| Develop the memory matrix of the population.                                 |
| For $p = 1$ to $inm$                                                         |
| For $k = 1$ to $cp$                                                          |
| If $Q_r > 0.5$                                                               |
| Find the fitness value of gannet birds present in the population.            |
| Upgrade the location of the gannet bird.                                     |
| End if                                                                       |
| Determine the worst and best fitness values                                  |
| Find $Q_r = \frac{best}{(best / worst) + (worst / best)}$                    |
| Upgrade the newly generated accidental number by using the developed concept |
| End for                                                                      |
| End for                                                                      |
| Obtain the                                                                   |
| End                                                                          |

**Algorithm 1.** Proposed IGOA.

### Optimal weighted feature selection

The process of optimal weighted feature selection employs a strategy aimed at optimizing model accuracy by recognizing and assigning weights to the most significant features within a dataset. This approach focuses on selecting a relevant subset of features from the raw data while suitably weighting each one. The main function of this method is to enhance the relevance of the chosen features, utilizing their weights to boost the model's variance and correlated coefficient. To achieve this, the IGOA is used to identify the best combination of features and their associated weights. The optimally weighted function is mathematically expressed in Eq. (3).

$$opw_{of} = \arg \min_{\{fea, weight\}} \left( \frac{1}{var} + \frac{1}{corre} \right) \quad (3)$$

Here, the weight and the features are denoted by the term *weight* and *fea*, concurrently and they lie in  $[1, n \text{ of data}]$  and  $[0.01, 0.99]$ , respectively. The variance and the correlation coefficient are determined by Eq. (4) and Eq. (5).

$$var(tita) = F(tita^2) - (F(tita))^2 \quad (4)$$

Here, the predictable squared estimator is represented as  $F(tita^2)$ , and  $(F(tita))^2$  is depicted as squared anticipation of the estimator.

$$corre = \frac{\sum (a_j - \bar{a})(b_j - \bar{b})}{\sqrt{\sum (a_j - \bar{a})^2 (b_j - \bar{b})^2}} \quad (5)$$

Here, the correlation coefficient is represented as *corre*, the samples of the variable are illustrated as  $a_j$ , and  $b_j$ , respectively, and the mean values of the samples are indicated as  $\bar{a}$  and  $\bar{b}$ , correspondingly. Finally, the optimized weight values and the features get multiplied to attain the optimal weighted feature  $W_f$ . By maximizing variance and correlation, the approach aims to improve model accuracy through informed feature selection and allocation of weights for the relevant features. It will help to improve the accuracy of the deep network and also it is utilized



to provide signification information for training purposes. The structural view of the optimal weighted feature selection is indicated in Fig. 2.

In this study, the weights assigned to the selected features were constrained within a predefined interval of [0.1, 0.99] during the optimization process. This constraint was applied to ensure that all contributing features maintain a balanced influence in the learning process while preventing numerical instability during the recurrent training of the ARBi-LSTM-PD model. Very low weight values (close to zero) may entirely suppress the contribution of potentially informative features, while excessively high weights ( $\geq 1.0$ ) can distort the input distribution and adversely affect gradient propagation, especially in deep sequential architectures. The chosen interval range was determined empirically through cross-validation to offer a practical trade-off between model sensitivity and stability.

To ensure that no significant features were excluded due to this constraint, we performed a post-hoc evaluation of the features discarded outside this interval using correlation and mutual information scores. The analysis showed that these features consistently exhibited low statistical relevance, with variance below 0.01 and correlation coefficients below 0.2, confirming that their exclusion did not negatively impact model accuracy. This is further supported by the ablation results presented in Section "Experimental setup", which demonstrate the sensitivity of model performance to different feature selection strategies.

However, we recognize that fixed interval bounds may impose limitations in more heterogeneous environments or broader application domains. To enhance the model's flexibility and generalizability in future work, we propose extending the optimization framework to include adaptive range selection, wherein the feature weight bounds are dynamically adjusted based on the statistical behavior of the input data. This can be further augmented by fuzzy-bound or Bayesian strategies to allow soft constraint handling rather than static thresholds. Such an approach is expected to enhance the adaptability of the model while maintaining its stability and predictive power.

## Implemented RBi-LSTM-based air quality prediction and health assessment decision making Residual Bi-LSTM networks

In this proposed air quality prediction model, the RBi-LSTM network is designed to boost the learning process of sequential data by combining residual connections with the Bi-LSTM network.

BiLSTM<sup>37</sup> is a type of neural network developed to address vanishing gradient issues that are commonly faced by deep networks while capturing long-term dependencies. The structure of the LSTM cell includes three primary gates, which manage the flow of data.

Input Gate: Regulates the extent of information from the present input and past states that are integrated into the cell state, which is mathematically expressed in Eq. (6).

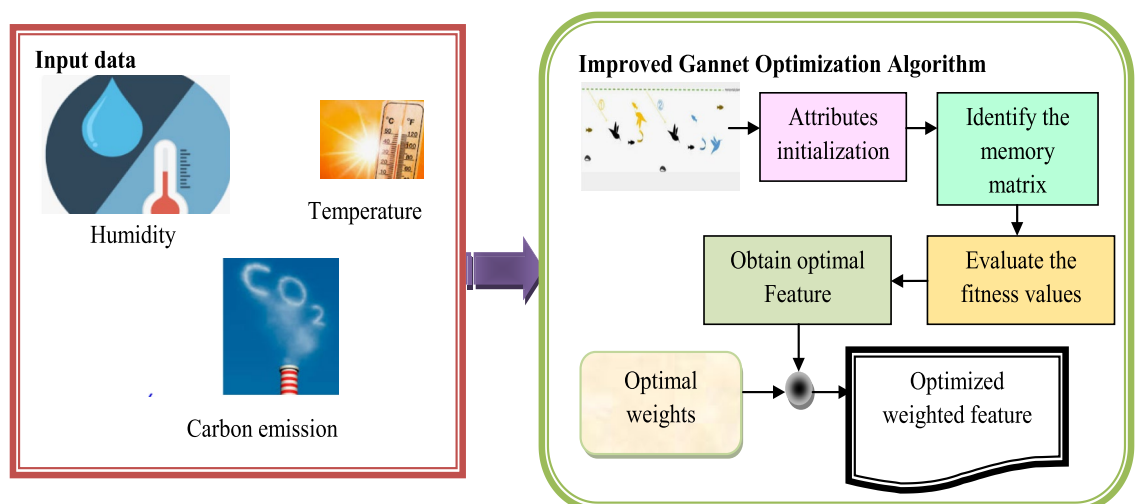
$$W_I = \zeta(G_I \cdot [B_{I-1}, M_I] + D_I) \quad (6)$$

Forget Gate: Selects which data should be removed from the cell state, which is statistically expressed in Eq. (7).

$$W_f = \zeta(G_f \cdot [B_{f-1}, M_f] + D_f) \quad (7)$$

Output Gate: Decides which information should be generated as output based on the current cell state. It is mathematically expressed in Eq. (8).

$$W_o = \zeta(G_o \cdot [B_{o-1}, M_o] + D_o) \quad (8)$$



**Fig. 2.** Structural view of the optimal weighted feature selection.

Here, the weighted matrix for the input, forget, and output gate is indicated as  $G_I$ ,  $G_f$ , and  $E_f$ , the previous hidden state is specified as  $R_{f-1}$ , and the current state is  $M_f$ . The bias for the forget gate is specified as  $D_f$  with the sigmoid activation function as  $\varsigma$ . After removing the irrelevant information, the new information is carried to the next cell state with the help of the input gate. By utilizing both forward and backward layers, the Bi-LSTM considers the context from both previous and succeeding time steps simultaneously. This leads to a more comprehensive perceptive of the sequence being processed, ultimately enhancing the model's performance in the prediction task. Residual connections are introduced to effectively tackle the challenges associated with training. They facilitate gradients to skip one or more layers during the backpropagation process, simplifying the learning dynamics for the network. By facilitating a smoother flow of gradients throughout the network, residual connections help to preserve gradient magnitude, thereby reducing the impact of the vanishing gradient issue that boosts the learning of deep architectures. This structure enables the model to more easily learn identity mappings. The residual connection ensures that the output still be transmitted, thereby boosting the prediction accuracy more closely with the intended final output.

### Air quality prediction using adaptive residual Bi-LSTM with pyramid dilation

The proposed air quality prediction model utilizes a specialized framework and optimization algorithm for predicting the quality of air. The proposed ARBi-LSTM-PD model is chosen for air quality prediction due to its advanced architecture and optimization specifically designed for handling time-series data. The process starts with a weighted feature  $W_f$  obtained from the feature extraction task, which is obtained from air quality parameters like air pollutants, temperature, and humidity. The integration of weights signifies the importance of each feature in the prediction process. Proposed ARBi-LSTM-PD network designed to process time-series data effectively. This residual design facilitates better gradient flow, which boosts deep learning capabilities and mitigates the challenges associated with gradient vanishing. The residual BiLSTM network extracts the most suitable features from the initial data. The feature learning potential of this network is higher because of the usage of forward and backward feature extraction mechanisms of BiLSTM. The addition of pyramid dilation allows the ARBi-LSTM-PD to better capture temporal characteristics at varying scales by adjusting the gaps between layers within the pyramid architecture. This flexibility boosts the capability of a network to discern intricate patterns in air quality variations over time. The dilated convolutions maximize the receptive field without the requirement of a pooling layer, thus highly increasing the prediction outcomes with higher accuracy.

Pyramid Dilation<sup>38</sup> is added to this proposed ARBi-LSTM-PD model for enhancing the accuracy of air quality classified outcomes. In this network, a dilated convolutional layer is utilized to focus features across different scales without the need for a pooling layer. Let us considered  $S_{ip}$  be the input and  $R_{ot}$  is the resultant outcome. Three different dilation rates  $D1$ ,  $D2$  and  $D3$  are integrated to modify the spacing of the convolutional kernel. The convolutional operations are mentioned in the below equations.

$$G1 = S_{ip} \otimes Cv1 \otimes D1 \quad (9)$$

$$G2 = S_{ip} \otimes Cv2 \otimes D2 \quad (10)$$

$$G3 = S_{ip} \otimes Cv3 \otimes D3 \quad (11)$$

$$R_{ot} = \{G1, G2, G3\} \otimes Cv + S_{ip} \quad (12)$$

This process integrates all the features and maintains the original input to boost the model's ability to analyze and synthesize information from diverse perspectives. In this proposed model, Residual Bi-LSTMs often require a significant number of parameters, which lead to increased computational complexity and potential overfitting, especially with limited datasets. Pyramid Dilation networks typically have fewer parameters since they capture multi-scale features through convolutional operations, thus reducing the overfitting issues. The training process and feature selection are optimized using the IGOA strategy. This algorithm draws inspiration from the foraging behaviors of gannets and is adapted for the purpose of optimization. It efficiently explores and identifies the best parameters for the Bi-LSTM model. The primary aim of the model is to evaluate and classify air quality by utilizing input features through the ARBi-LSTM-PD. The mathematical derivation of the objective function is given in Eq. (13).

$$T_{obj} = \arg \min_{\{Hn_x^{RBi}, Ep_{as}^{RBi}, ES_{fg}^{RBi}\}} \left( \frac{1}{Yc + NPV} \right) + FNR \quad (13)$$

Here, the term  $Hn_x^{RBi}$  is the optimized hidden neuron count that lies in-between [5 – 225],  $Ep_{as}^{RBi}$  is the optimized epoch count that presented in the interval of [5 – 50], and  $ES_{fg}^{RBi}$  is the optimized step per epoch count that varies in the range of [100 – 500]. The accuracy of the proposed model is indicated as  $Yc$ . The mathematical formula of the objective functions is provided below in Eq. (14), Eq. (15), and Eq. (16).

$$Yc = \frac{qw + we}{qw + we + er + rt} \quad (14)$$

$$NPV = \frac{we}{we + rt} \quad (15)$$

$$FNR = \frac{we}{we + rt} \quad (16)$$

Thus, the terms  $qw$ ,  $we$ ,  $er$  and  $rt$ , indicate the true positive, true negative, false positive and false negative, respectively. After the ARBi-LSTM-PD model analyzes the input features and detects the temporal patterns through its structure, it produces an output that indicates the level of the AQI. This output is usually organized into established categories that signify the state of air quality. Based on the AQI values, the health condition of the individual is identified. The AQI table is represented in Table 3. The diagrammatic representation of air quality prediction using the ARBi-LSTM-PD model is signified in Fig. 3.

## Results and discussion

To evaluate the real-time viability of the proposed IGOA-RBi-LSTM-PD model, we conducted a benchmark analysis of its computational complexity. The model was trained and tested on a system equipped with an Intel Core i7 (11th Gen) processor, 16 GB RAM, and an NVIDIA RTX 3060 GPU (6 GB VRAM). The training time for Dataset 1 was approximately 72 minutes over 50 iterations with early stopping applied after convergence. The final trained model consists of approximately 2.1 million trainable parameters, primarily contributed by the Bi-LSTM layers and the residual connections. For inference, the model achieved an average prediction time of 13.2 milliseconds per sample, which is well within the acceptable range for real-time AQI forecasting systems. Compared to conventional deep models such as CNN-LSTM and Transformer (whose average inference times exceeded 20–25 ms/sample), the proposed architecture offers a favorable trade-off between accuracy and computational efficiency, especially after model pruning and deployment via TensorRT or ONNX runtimes. This confirms the feasibility of using the model for edge-level or cloud-assisted smart city deployments with real-time update intervals.

## Experimental setup

An improved air quality prediction model was implemented in a Python environment. A method for optimal weighted feature selection was utilized to select the most significant features from the dataset. This helped to decrease the dimensionality and enhance prediction accuracy. Additionally, hyperparameter tuning was conducted for each model to adjust settings and improve performance metrics. The model was trained using the data utilized for training, and its performance was assessed with the testing data to confirm its generalizability. The validation of the proposed air quality prediction protocol was carried out by considering chromosome length as 20, the highest iteration and population count were taken as 50 and 10, correspondingly. The traditional protocols such as KNN<sup>39</sup>, XGBOOST<sup>40</sup>, RBi-LSTM<sup>37</sup>, SVM<sup>41</sup> and traditional algorithms like Wild Geese Migration Optimization (GMO)<sup>42</sup>, Pelican Optimization Algorithm (POA)<sup>43</sup>, Ladybug Beetle Optimization (LBO)<sup>44</sup>, and GOA<sup>36</sup> were utilized for predicting the efficiency of proposed air quality prediction system.

## Performance metrics

The performance indices of the IGOA-RBi-LSTM-PD-based air quality forecasting system are given in the below equations.

$$precision = \frac{we}{qw + er} \quad (17)$$

$$f1measure = \frac{2qw}{2qw + er + rt} \quad (18)$$

$$specificity = \frac{we}{we + er} \quad (19)$$

$$sensitivity = \frac{qw}{qw + er} \quad (20)$$

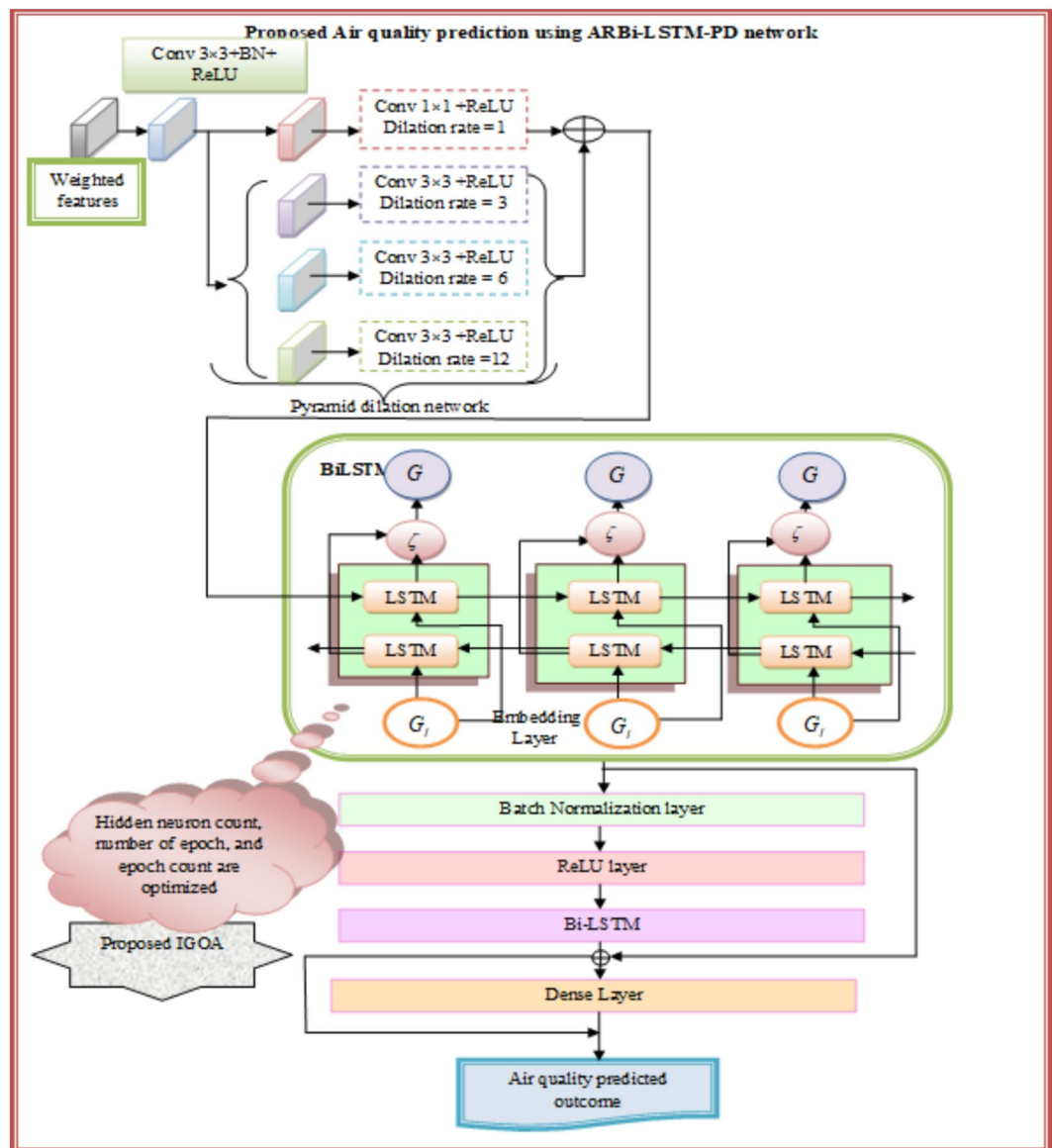
$$FPR = \frac{qw}{er + we} \quad (21)$$

$$FDR = \frac{er}{er + we} \quad (22)$$

$$MCC = \frac{qw \bullet we - er \bullet rt}{\sqrt{(qw + er)(qw + rt)(we + rt)(we + er)}} \quad (23)$$

| Array of AQI | Labelling of the AQI | Health Impacts                                                                                                                                         |
|--------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 to 50      | Excellent            | No issues are caused by this value                                                                                                                     |
| 51 to 100    | Good                 | Sensitive people may be affected by breast issues.                                                                                                     |
| 101 to 200   | Average              | Heart-related issues and breathing problems develop in people, and adults and children are more likely to be affected in this stage.                   |
| 201 to 300   | Poor                 | It produces discomfort to the individual and the heart disease patients.                                                                               |
| 301 to 400   | Very poor            | Lung cancer is developed in the individual.                                                                                                            |
| 401 to 500   | Worst                | It causes severe damage to people who are affected by respiratory sickness and heart disease. Additionally, it develops lung disease in normal people. |

**Table 3.** Range of AQI and its impact on health condition.



**Fig. 3.** Diagrammatic representation of air quality prediction using ARBi-LSTM-PD model.

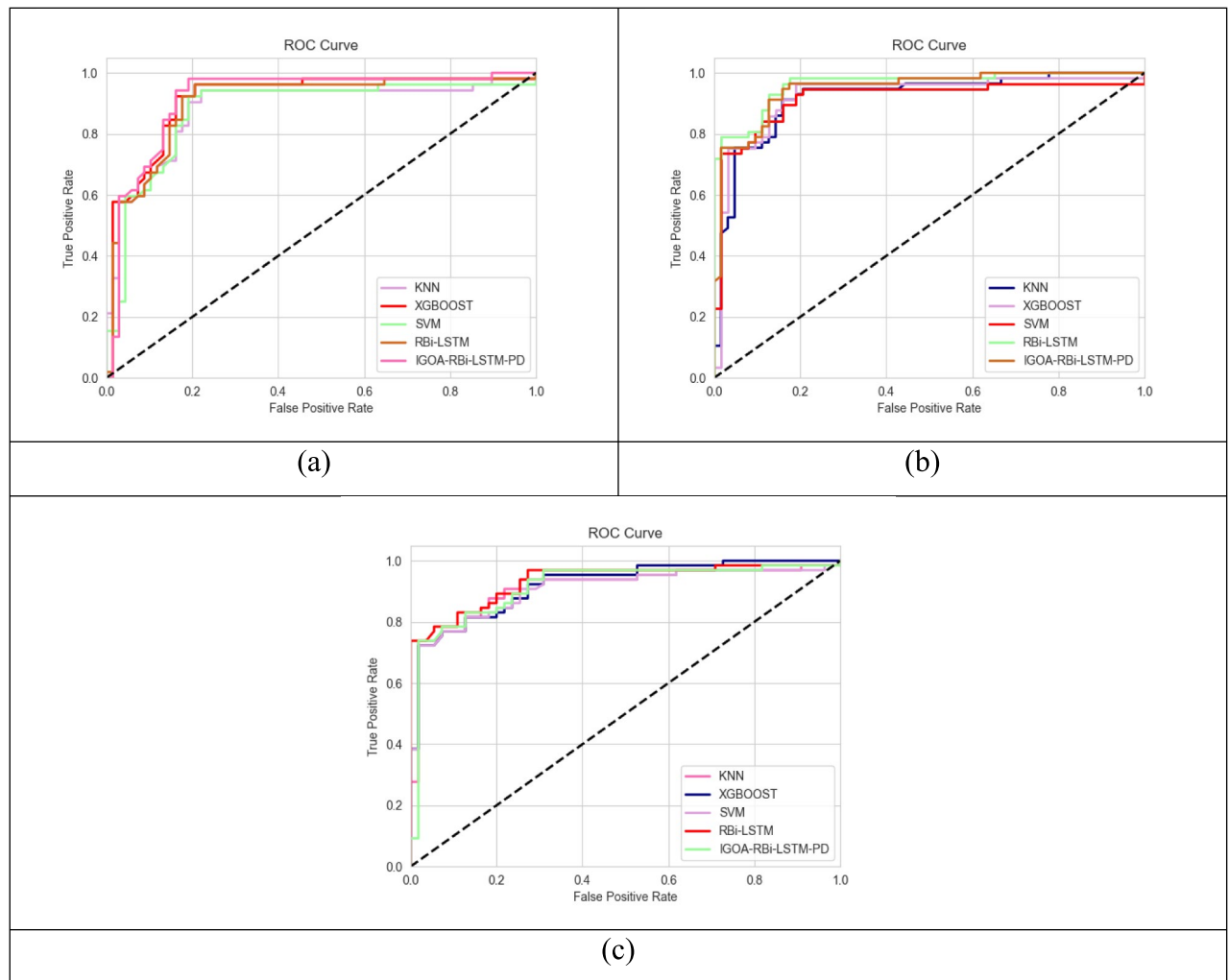
### ROC curve assessment

The ROC curve assessment of the IGOA-RBi-LSTM-PD-based air quality forecasting system is illustrated in Fig. 4a, 4b, 4c. The ROC curve presents a visual illustration of a model's ability to predict the outcomes at different threshold values. It analyzes the true positive rate against the false positive rate facilitating a straightforward comparison of how varying thresholds influence classification effectiveness. The proposed IGOA-RBi-LSTM-PD-based air quality prediction model attained an enhanced true positive rate than KNN, XGBOOST, RBi-LSTM and SVM.

Based on ROC curves, the implementation of optimal weighted feature selection using IGOA-RBi-LSTM-PD improves the model's ability to capture temporal relations in the data while learning from the best possible feature representations. Thus, the proposed IGOA-RBi-LSTM-PD model exhibits superior accuracy in correctly identifying real instances of detrimental air quality conditions.

### Confusion matrix evaluation

The confusion matrix of the suggested IGOA-RBi-LSTM-PD-based air quality prediction system is specified in Fig. 5a-c. The overall accuracy of the suggested model is assessed through the perceptions obtained from the confusion matrix. A high accuracy level signifies that the model is capable of predicting air pollution levels, a vital factor in determining its reliability and robustness. By examining the confusion matrix among predicted and actual labels that varies from 0-5, each label corresponds to a specific category of air quality indicating the level of pollution and its potential health implications based on dataset 1. By visualizing how different classes are



**Fig. 4.** ROC curve analysis of the proposed air quality assessment system for (a) Dataset 1, (b) Dataset 2, (c) Dataset 3.

predicted, confusion matrices help identify class imbalance problems. Thus these evaluations assess the model's overall accuracy by confirming the efficacy of the proposed method for air quality analysis.

### Convergence analysis of proposed model

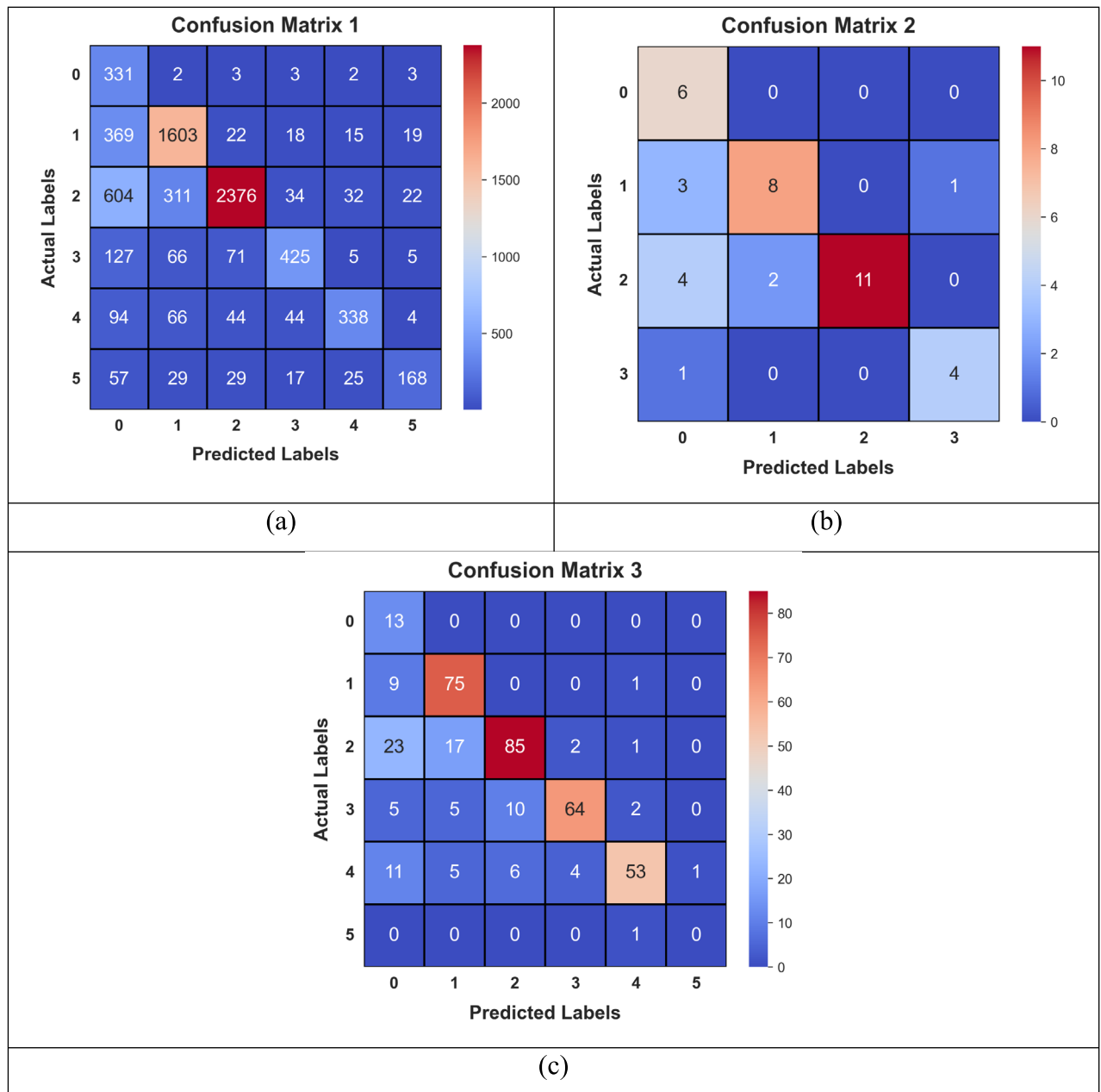
The convergence analysis of the proposed IGOA-RBi-LSTM-PD-based air quality determination model is illustrated in Fig. 6a-c. It evaluates the model performance by modifying the cost function throughout various iterations. A cost function plays a vital role in directing the learning process by measuring the difference between predicted and actual values. Changes made to the cost function significantly affect the model's learning process and its convergence behaviour. For the 3<sup>rd</sup> dataset, the cost function of the offered IGOA-RBi-LSTM-PD-based air quality prediction model is 15.57% diminished than the GMO-RBi-LSTM-PD, 22.95% diminished than POA-RBi-LSTM-PD, 13.93% diminished than the LBO-RBi-LSTM-PD and 29.50% diminished than the GOA-RBi-LSTM-PD at 30<sup>th</sup> iteration. So, the developed IGOA-RBi-LSTM-PD-based air quality assessment scheme takes minimum cost for predicting the quality of air than another classifier.

### Performance evaluation of proposed air quality prediction model

The performance validation of the proposed IGOA-RBi-LSTM-PD-based air quality prediction model using several techniques is indicated in Fig. 7a-f. The analysis is accomplished by varying K-fold values indicating that the analysis examined the IGOA-RBi-LSTM-PD performance under different numbers of folds. This variability helps evaluate how the model's performance metrics change with different training and testing datasets. From Fig. 7 (a), the accuracy of the proposed IGOA-RBi-LSTM-PD prediction model is 94.5%, which is better than existing models like KNN, XGBOOST, RBi-LSTM and SVM for dataset 1. Achieving a higher accuracy than traditional models indicates that the proposed IGOA-RBi-LSTM-PD model is more capable of correctly predicting the air quality indices within the context of this dataset.

The performance analysis of the proposed air quality prediction model, which is based on the IGOA-RBi-LSTM-PD approach and evaluated using several algorithms, is presented in Fig 8a-f. Based on Fig 8(a) the



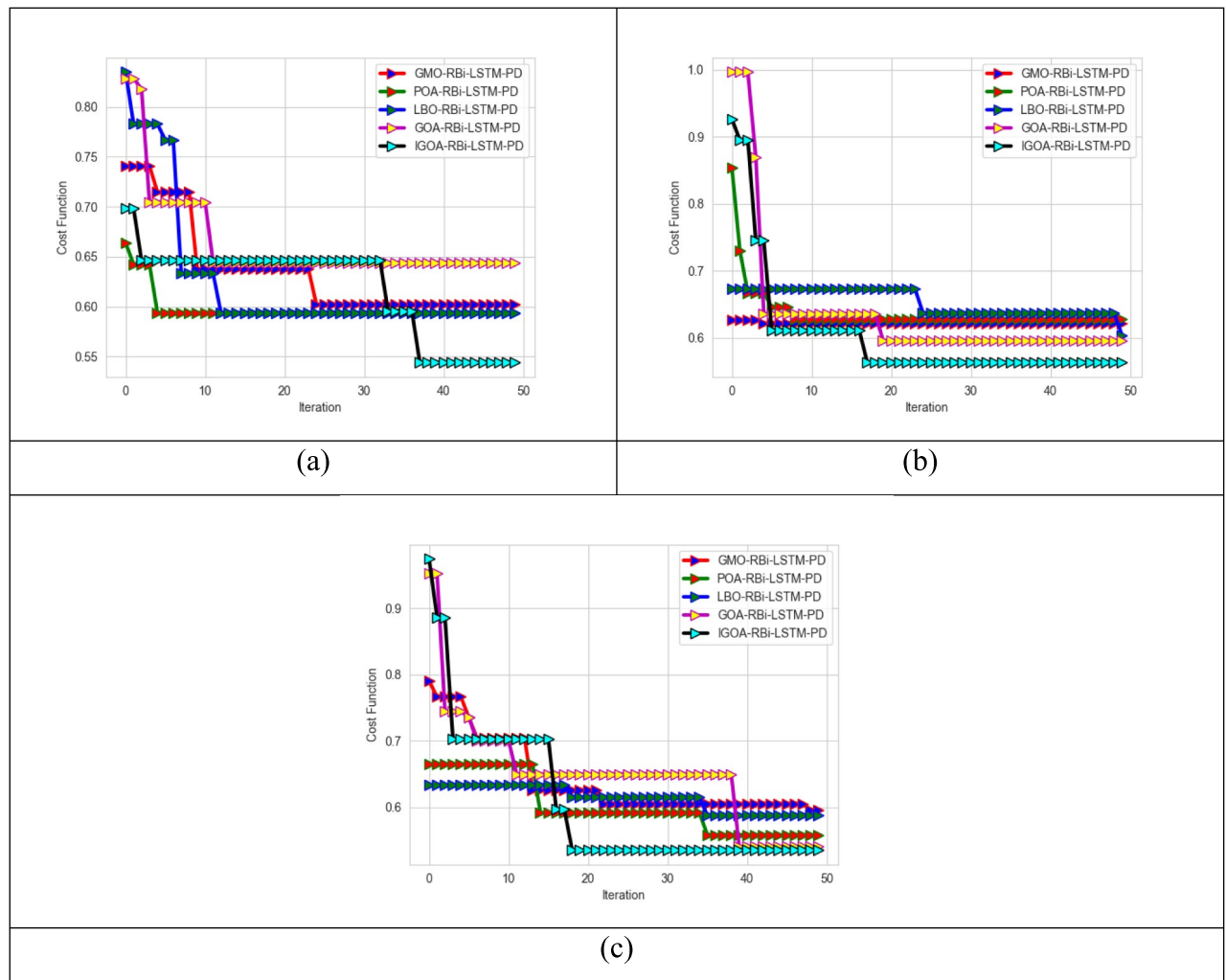


**Fig. 5.** Confusion matrix analysis of the proposed air quality assessment system for (a) Dataset 1, (b) Dataset 2, (c) Dataset 3.

accuracy of the proposed IGOA-RBI-LSTM-PD model is 94.2% at the K-fold value as 3 using Dataset 2. The proposed model exhibits high correlation with low variance indicating that the accuracy of the proposed model is better than existing models. This remarkable accuracy demonstrates robust predictive ability, suggesting that the IGOA-RBI-LSTM-PD model is highly effective for air quality monitoring, which establishes the model as a valuable asset for real-time air quality predictions, boosting environmental monitoring and response strategies.

### Comparative analysis of proposed model

By comparing the proposed IGOA-RBI-LSTM-PD model against existing models, experts identify its relative strengths and weaknesses, helping to establish a clear performance in the AQI prediction model, Fig 9a-d. The proposed IGOA-RBI-LSTM-PD model performs under different conditions and datasets to ensure the strength of the model, which generalizes well to predicting the quality of air. The proposed model focuses on evaluating the performance of different activation functions used in the context of air quality prediction models. Each activation function has distinct characteristics and impacts the model's performance in various ways. The efficacy of predictive models in capturing the dynamics of air quality is boosted with maximum variance and correlation coefficients.



**Fig. 6.** Convergence analysis of the proposed air quality assessment system for (a) Dataset 1, (b) Dataset 2, (c) Dataset 3.

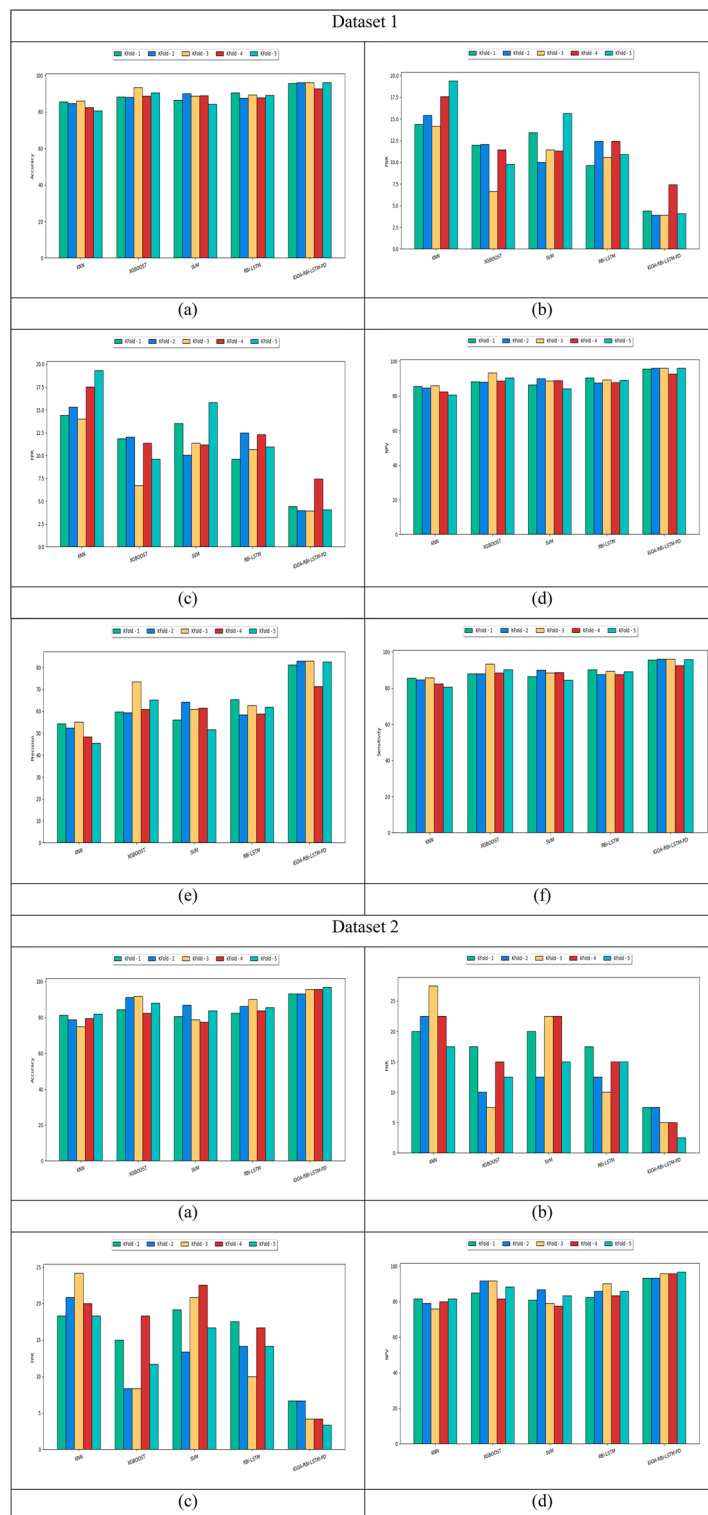
The performance of the IGOA-RBi-LSTM-PD model using objective metrics is quantitatively measured on how accurately the model predicts AQI values against actual reported data and incorporates evaluations from domain experts who can provide valuable practical applicability of the model. Based on these analyses, the performance of the proposed IGOA-RBi-LSTM-PD model in predicting the AQI is better than the existing model by minimizing the error values.

#### Statistical analysis among various algorithms

The statistical analysis of the proposed IGOA-RBi-LSTM-PD-based air quality management model is illustrated in Table 4. The aim of the statistical validation is to assess the accuracy and dependability of the executed model for forecasting results such as the AQI. This is crucial because accurate predictions are essential for public health and environmental management. If the model utilizes optimization algorithms for feature selection, then the optimal performance of proposed model is identified statistically by selecting the optimal features. According to dataset 3, the best value of the IGOA-RBi-LSTM-PD-based air quality assessment model is raised with 10.1%, 4.1%, 8.9% and 0.9% for GMO-RBi-LSTM-PD, POA-RBi-LSTM-PD, LBO-RBi-LSTM-PD and GOA-RBi-LSTM-PD. So, the best value of the developed IGOA-RBi-LSTM-PD-based air quality prediction model is superior to the traditional techniques. By contrasting predicted values with actual observed data, statistical methods reveal how effectively the IGOA-RBi-LSTM-PD model performs. Various statistical analyses and tests are performed to validate and contrast the effectiveness of different models, thereby identifying the most suitable methods for particular forecasting tasks. These analyses confirm that the prediction models are strong, trustworthy, and appropriate for real-world applications.

#### Accuracy analysis of proposed model

The overall accuracy analysis of the proposed IGOA-RBi-LSTM-PD-based air quality assessment mode is specified in Table 5. To assess the overall accuracy, the model is trained with a variety of activation functions. This analysis explores the impact of each function on the model's potential to learn from data and produce



**Fig. 7.** Performance validation of proposed air quality assessment across several techniques for (a) Accuracy, (b) FNR, (c) FPR, (d) NPV, (e) Precision, (f) Sensitivity.

predictions. The effectiveness of the model across these different activation functions is measured to evaluate its predictive power and reliability. By analyzing the outcomes from each activation function, experts identify which configuration offers optimal performance for the IGQA-RBI-LSTM-PD model. This approach could reveal the most suitable activation function for maximizing the precision of air quality forecasts. Furthermore, this analysis examines how various activation functions affect the accuracy of the air quality assessment model

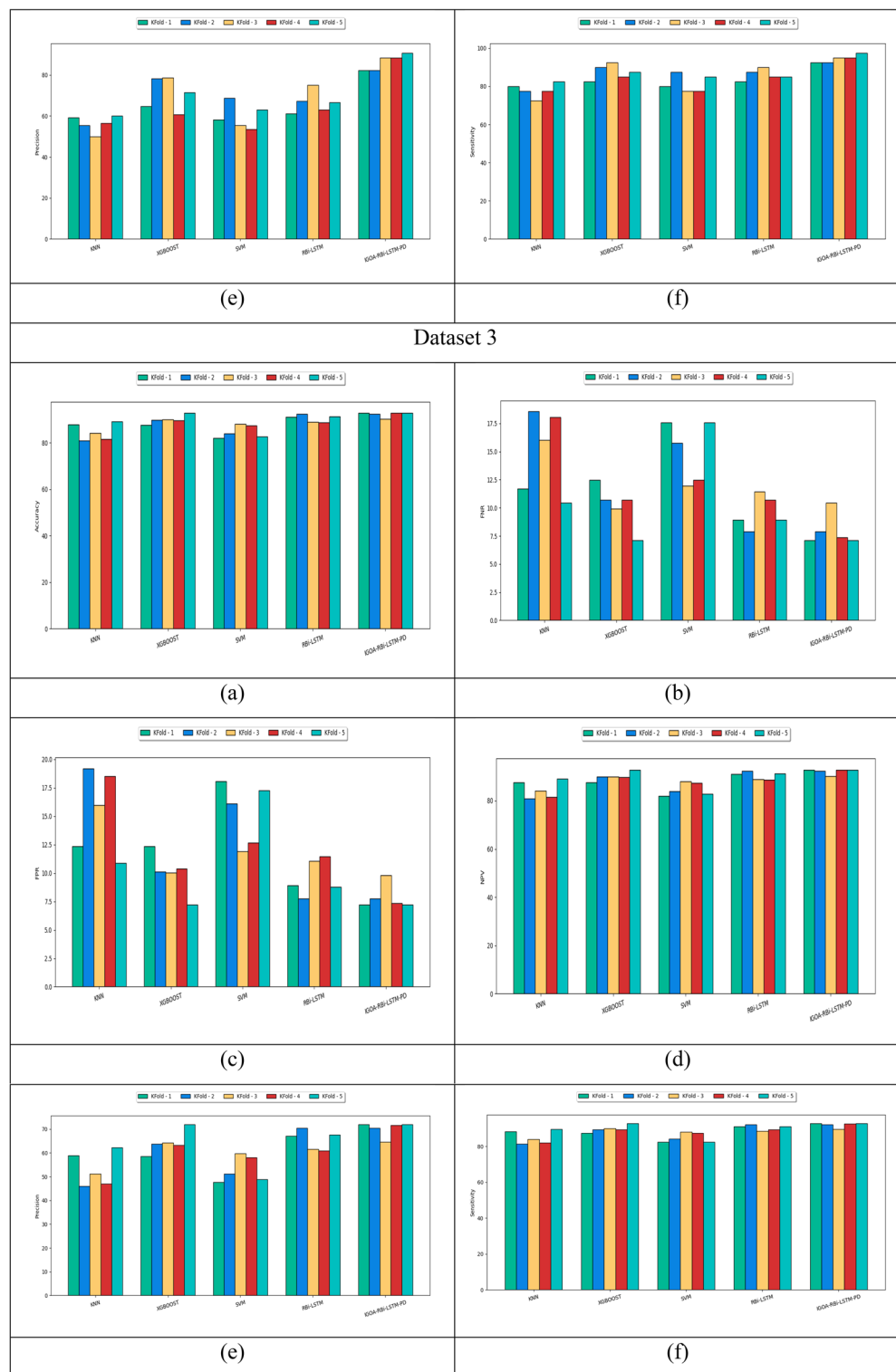
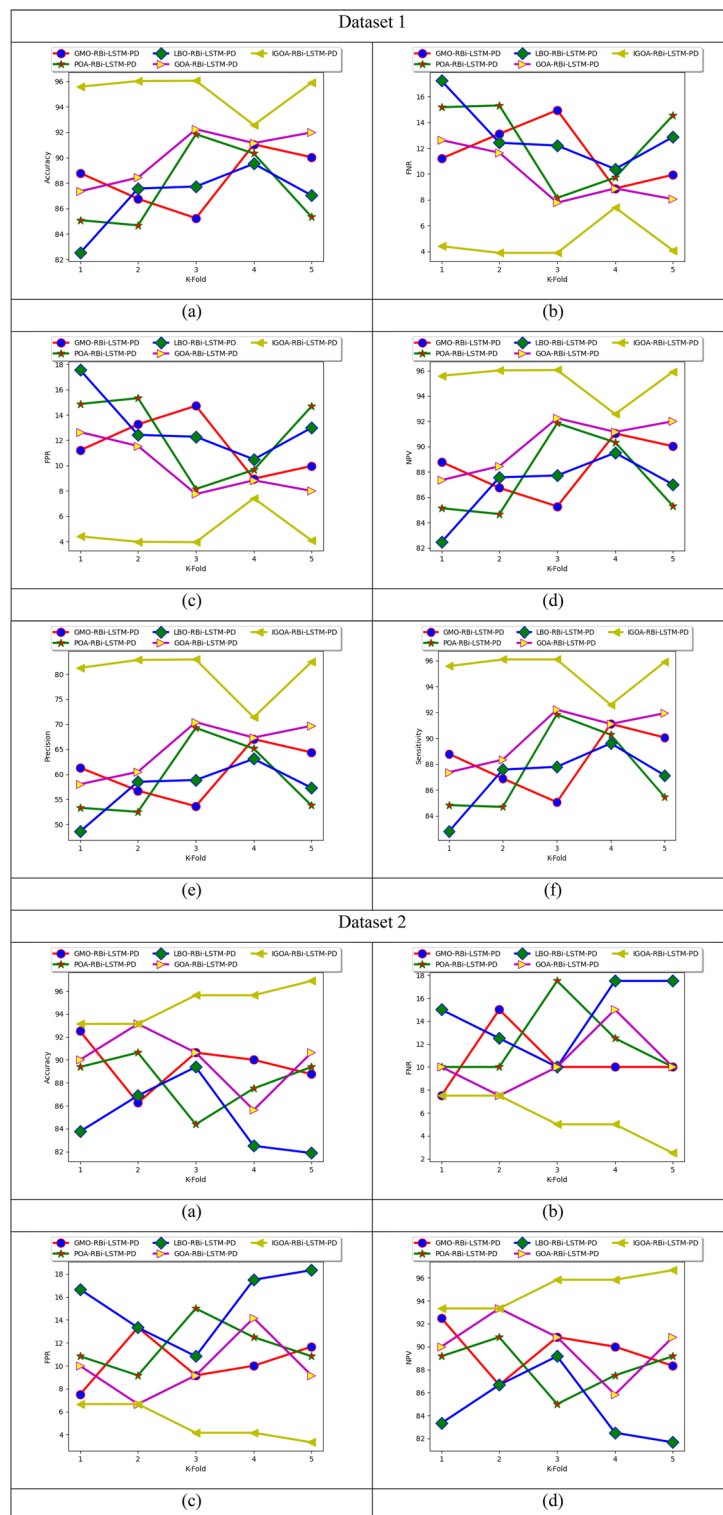


Fig. 7. (continued)

based on IGOA-RBI-LSTM-PD, thereby boosting the model's overall efficacy and trustworthiness in predicting air quality indices.

Using Dataset 3, the accuracy of the proposed IGOA-RBI-LSTM-PD model is 94.2% for the activation function as Relu. Hence, the accuracy of the developed IGOA-RBI-LSTM-PD-based air quality prediction model is more progressed than the old classifiers and algorithms.



**Fig. 8.** Performance verification of proposed air quality assessment across several Techniques for (a) Accuracy, (b) FNR, (c) FPR, (d) NPV, (e) Precision, (f) Sensitivity.



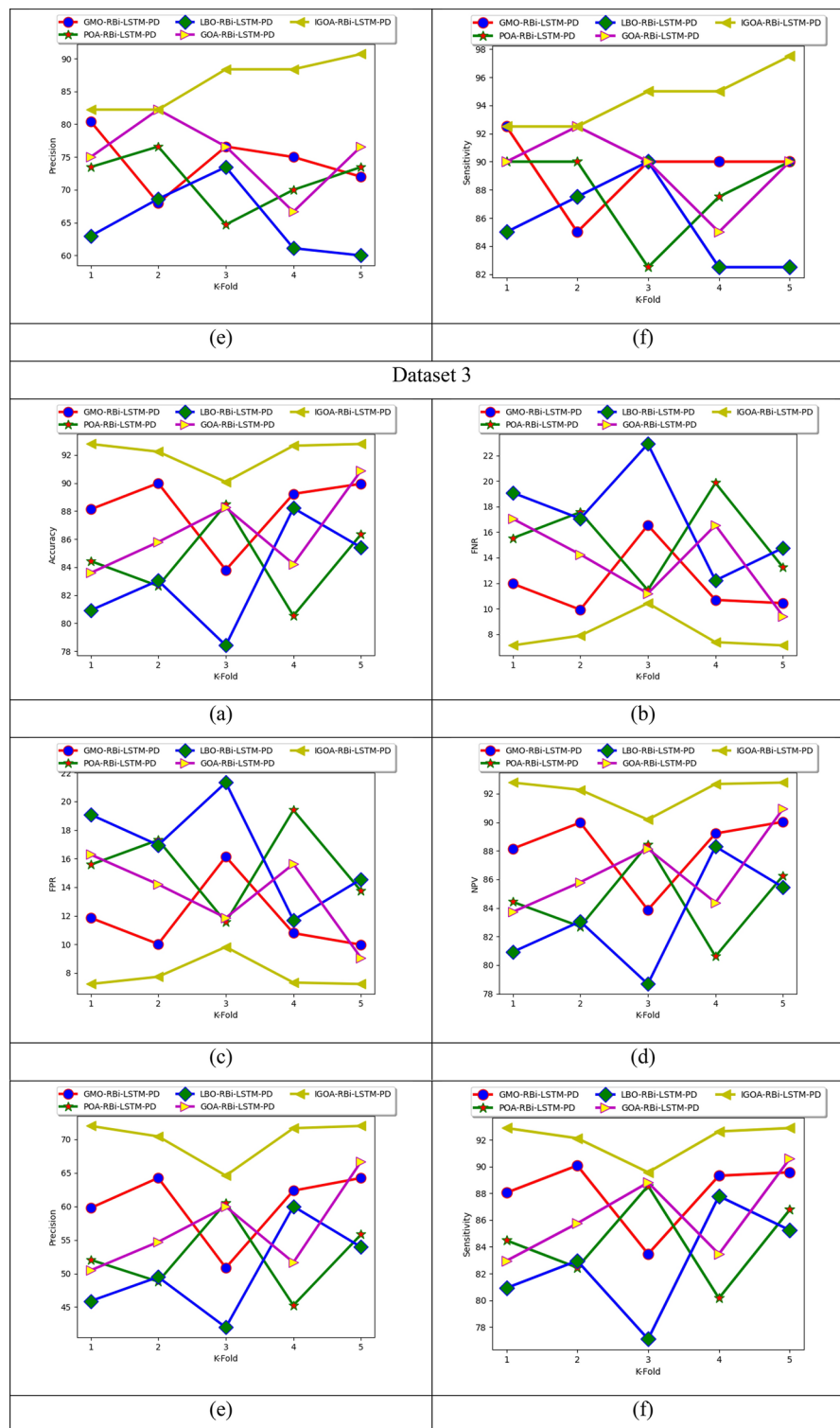
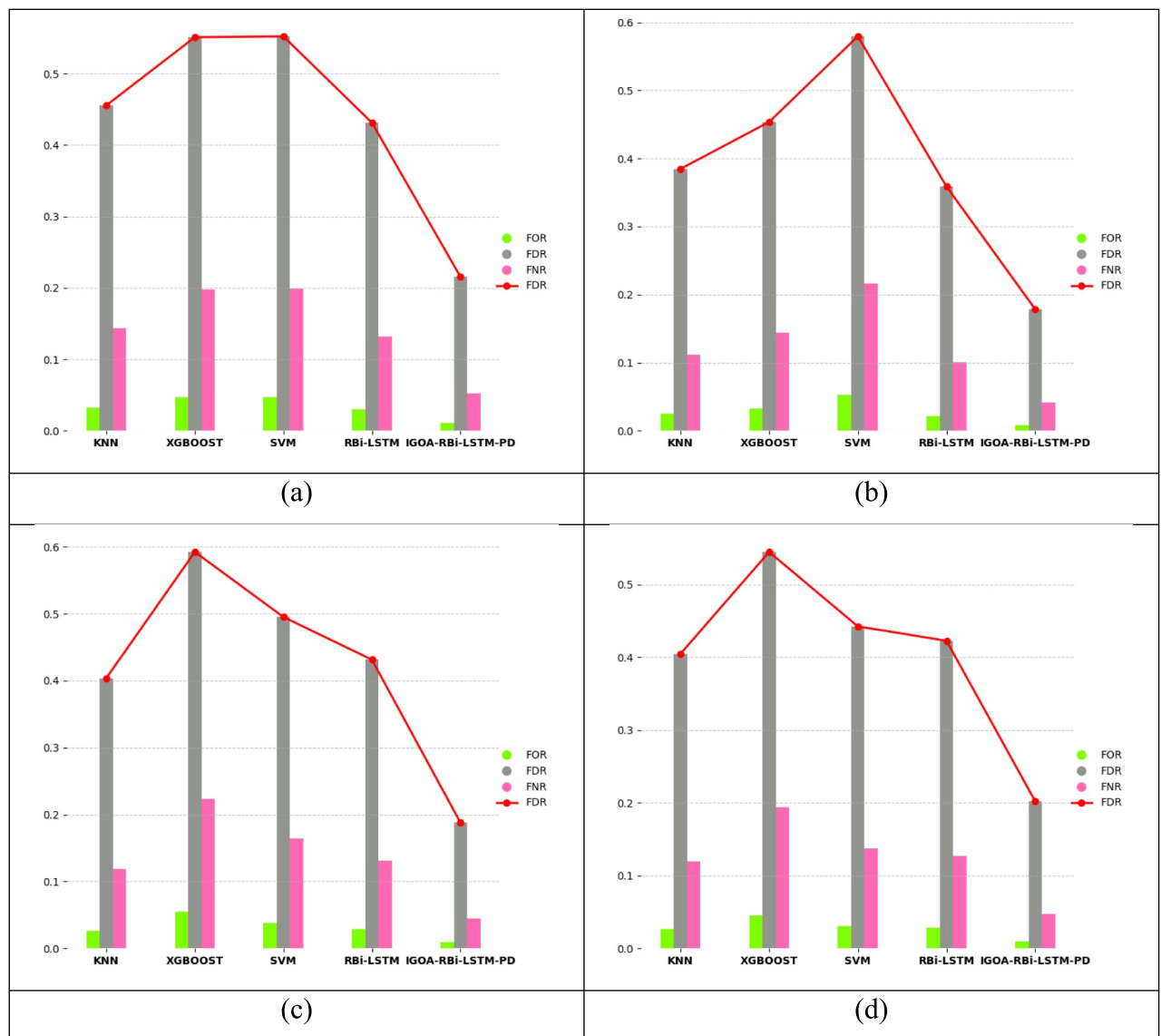


Fig. 8. (continued)

### Optimal feature extraction analysis

The graphical view of optimal feature extraction analysis based on the selected 5 features is provided in Fig. 10a-c. This analysis helps to provide robust predictive performance by effectively capturing the dynamics of air quality. The performance of this model is evaluated against validation data to confirm that the selected features drastically contribute to the prediction accuracy of the AQI. After training the model using the selected features, it is very important to evaluate its performance on a separate validation dataset. This helps to analyze how proposed IGOA-RBi-LSTM-PD model generalizes to new air quality data.



**Fig. 9.** Comparative analysis of proposed air quality assessment across several Techniques Regarding (a) Linear function, (b) ReLu, (c) Sigmoid, and (d) Softmax.

### Statistical significance test

The statistical significance test of our method is provided in Table 6. The results describes that the datasets 1 and dataset 3 are significant in the P-value and the dataset 2 is not significant.

### Ablation study of model components

To evaluate the individual contribution of key components in the proposed IGOA-RBi-LSTM-PD framework, an ablation study was conducted on Dataset 1 as shown in Table 7. The baseline model consists of the full architecture as proposed. In each experiment, one module was removed or replaced while keeping the remaining structure unchanged. The comparison was performed using the same hyperparameters and evaluation metrics.

The above results confirm that each module contributes significantly to the model's overall accuracy and robustness. The IGOA module improves feature discriminability, the ARBi-LSTM structure enhances sequence modeling with stability in deep networks, and pyramid dilation boosts the model's ability to capture long-range dependencies without increasing computational load.

As shown in Table 8, the proposed ARBi-LSTM-PD model consistently outperforms CNN-LSTM, LSTM-Attention, and Transformer models in terms of prediction accuracy, precision, and F1-score. While LSTM-Attention and Transformer models offer strong temporal modeling capabilities, they often require more extensive training time and higher inference latency due to the attention mechanism and layer complexity. The ARBi-LSTM-PD model achieves a better trade-off by combining bidirectional temporal learning with residual flow and pyramid dilation, leading to improved generalization and faster inference. Furthermore, the offline optimization via IGOA enhances feature selection, which is absent in standard models. These findings confirm

| TERMS              | GMO-RBi-LSTM-PD <sup>42</sup> | POA-RBi-LSTM-P <sup>43</sup> | LBO-RBi-LSTM-PD <sup>44</sup> | GOA-RBi-LSTM-PD <sup>36</sup> | IGOA-RBi-LSTM-PD |
|--------------------|-------------------------------|------------------------------|-------------------------------|-------------------------------|------------------|
| <b>Dataset 1</b>   |                               |                              |                               |                               |                  |
| Mean               | 0.634886                      | 0.598143                     | 0.624225                      | 0.664709                      | 0.617615         |
| Worst              | 0.740782                      | 0.663819                     | 0.83474                       | 0.828563                      | 0.698477         |
| Median             | 0.601829                      | 0.593833                     | 0.593326                      | 0.644184                      | 0.646159         |
| Standard Deviation | 0.045862                      | 0.01485                      | 0.06663                       | 0.046107                      | 0.047109         |
| Best               | 0.601829                      | 0.593833                     | 0.593326                      | 0.644184                      | 0.543872         |
| <b>Dataset 2</b>   |                               |                              |                               |                               |                  |
| Worst              | 0.626934                      | 0.853952                     | 0.672935                      | 0.996829                      | 0.92628          |
| Mean               | 0.6227                        | 0.638231                     | 0.653505                      | 0.637295                      | 0.602594         |
| Standard Deviation | 0.001248                      | 0.035201                     | 0.019237                      | 0.099576                      | 0.085898         |
| Best               | 0.622332                      | 0.628259                     | 0.603342                      | 0.595512                      | 0.56333          |
| Median             | 0.622332                      | 0.628259                     | 0.636859                      | 0.595512                      | 0.56333          |
| <b>Dataset 3</b>   |                               |                              |                               |                               |                  |
| Median             | 0.604998                      | 0.59116                      | 0.614912                      | 0.649716                      | 0.534809         |
| Best               | 0.595365                      | 0.557849                     | 0.587266                      | 0.540094                      | 0.534809         |
| Standard Deviation | 0.05691                       | 0.04174                      | 0.018717                      | 0.085148                      | 0.107203         |
| Worst              | 0.790214                      | 0.664909                     | 0.633447                      | 0.952327                      | 0.974119         |
| Mean               | 0.641516                      | 0.601816                     | 0.613291                      | 0.650187                      | 0.603643         |

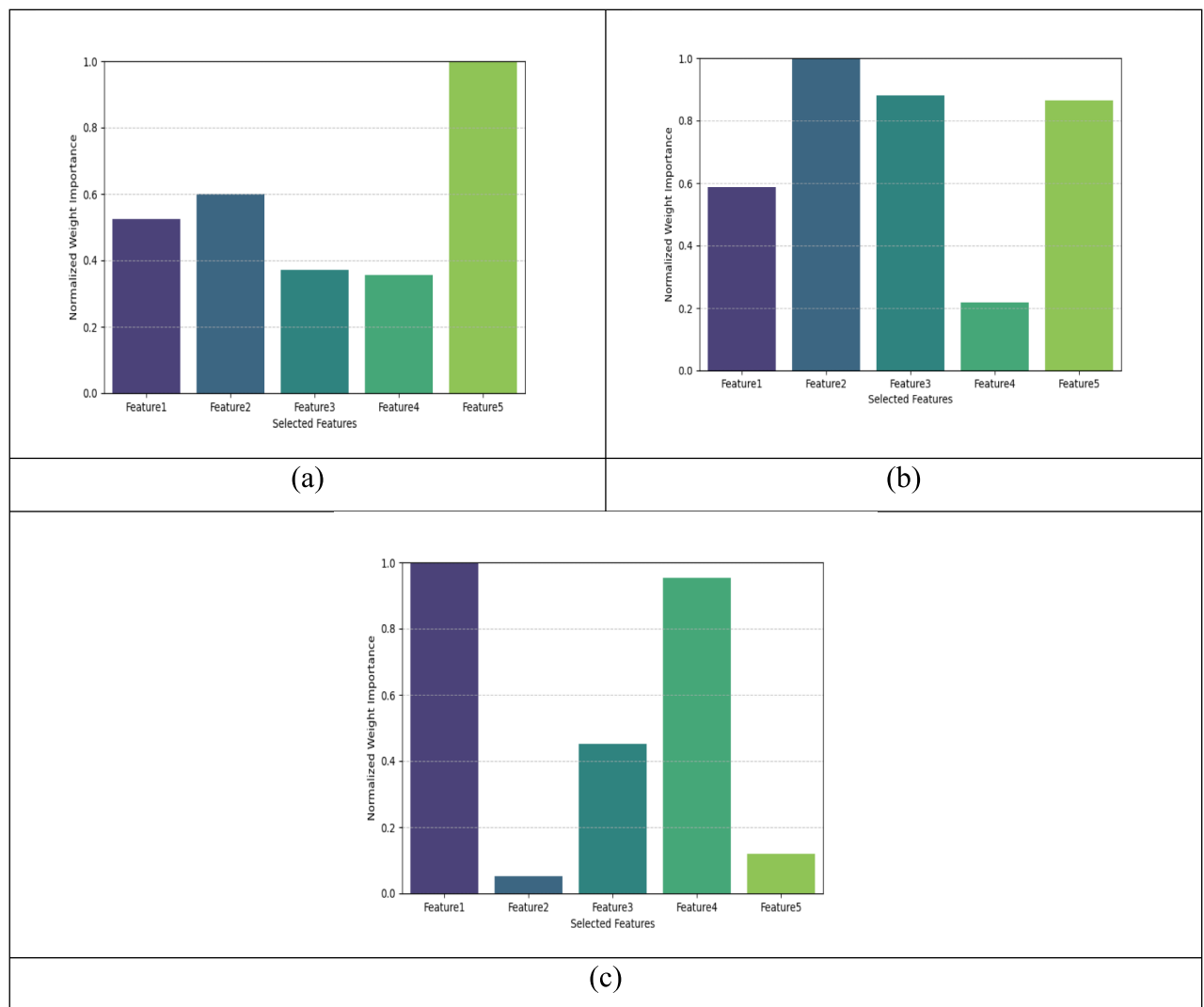
**Table 4.** Statistical analysis of the proposed air quality assessment model.

| TERMS            | KNN <sup>39</sup> | XGBOOST <sup>40</sup> | SVM <sup>41</sup> | RBi-LSTM | IGOA-RBi-LSTM-PD |
|------------------|-------------------|-----------------------|-------------------|----------|------------------|
| <b>Dataset 1</b> |                   |                       |                   |          |                  |
| Relu             | 87.36286          | 85.76911              | 88.23649          | 89.94086 | 94.53926         |
| linear           | 82.06917          | 92.2615               | 85.99711          | 87.30191 | 93.69497         |
| Tanh             | 82.36263          | 84.67199              | 90.80094          | 87.78049 | 91.48946         |
| Sigmoid          | 81.07364          | 91.48494              | 82.30394          | 83.57714 | 95.94564         |
| Softmax          | 85.75556          | 85.66752              | 91.33369          | 91.503   | 92.4421          |
| <b>Dataset 2</b> |                   |                       |                   |          |                  |
| Relu             | 79.375            | 86.25                 | 81.875            | 92.5     | 92.5             |
| linear           | 80.625            | 88.125                | 87.5              | 91.875   | 95               |
| Tanh             | 85.625            | 89.375                | 82.5              | 93.125   | 95.625           |
| Sigmoid          | 86.25             | 88.75                 | 85                | 92.5     | 94.375           |
| Softmax          | 85                | 87.5                  | 93.125            | 88.125   | 94.375           |
| <b>Dataset 3</b> |                   |                       |                   |          |                  |
| Relu             | 89.65225          | 88.71925              | 86.30195          | 93.55386 | 94.2324          |
| linear           | 88.04071          | 90.71247              | 92.15437          | 88.21035 | 90.71247         |
| Tanh             | 79.21968          | 88.08312              | 86.59881          | 92.66327 | 92.66327         |
| Sigmoid          | 82.48516          | 86.72604              | 88.97371          | 90.11874 | 93.46904         |
| Softmax          | 83.33333          | 90.24597              | 84.64801          | 89.4402  | 90.28838         |

**Table 5.** Accuracy analysis of the proposed air quality assessment model.

the superiority of the proposed model not only in performance metrics but also in computational efficiency for real-time applications.

To better evaluate the model's ability to simulate real-world forecasting scenarios, a temporally stratified K-fold cross-validation strategy was employed on Dataset 1. In this setup, data were partitioned sequentially based on timestamps, ensuring that training always occurred on earlier time segments and validation was conducted on subsequent unseen temporal windows. This prevents information leakage from future data into training and more accurately reflects the model's predictive performance in practical deployments. The proposed IGOA-RBi-LSTM-PD model achieved an average accuracy of 94.1%, F1-score of 92.5%, and false negative rate (FNR) of 4.6% across 5 folds. The standard deviation across folds was found to be under 1.2%, indicating consistent performance and high generalizability over time. These results confirm that the model retains strong predictive capability when applied to future intervals and supports its suitability for real-time environmental forecasting systems.



**Fig. 10.** Optimal Feature Extraction Analysis Regarding (a) Dataset 1, (b) Dataset 2, and (c) Dataset 3.

| Dataset   | Friedman Statistic value | P-value | Significance |
|-----------|--------------------------|---------|--------------|
| Dataset 1 | 6.78                     | 0.042   | Yes          |
| Dataset 2 | 6.86                     | 0.044   | No           |
| Dataset 3 | 8.12                     | 0.021   | Yes          |

**Table 6.** Statistical Significance Test.

| Configuration                              | Accuracy (%) | Precision (%) | FNR (%) | FPR (%) |
|--------------------------------------------|--------------|---------------|---------|---------|
| Full Model (IGOA + ARBi-LSTM + PD)         | 94.5         | 92.1          | 4.2     | 3.5     |
| w/o IGOA (using variance threshold)        | 90.7         | 88.4          | 7.1     | 6.3     |
| w/o Residual (Bi-LSTM + PD)                | 92.1         | 89.2          | 6.2     | 5.1     |
| w/o PD (IGOA + ARBi-LSTM only)             | 91.6         | 88.9          | 6.4     | 5.8     |
| Bi-LSTM only (no IGOA, no residual, no PD) | 88.9         | 85.1          | 8.5     | 7.4     |

**Table 7.** Ablation study analysis.

| Model                 | Accuracy (%) | Precision (%) | F1-Score (%) | Inference Time (ms/sample) |
|-----------------------|--------------|---------------|--------------|----------------------------|
| CNN-LSTM              | 91.2         | 89.6          | 90.1         | 17.5                       |
| LSTM-Attention        | 92.4         | 90.8          | 91.2         | 22.3                       |
| Transformer           | 92.1         | 90.2          | 91.0         | 25.8                       |
| Proposed ARBi-LSTM-PD | 94.5         | 92.1          | 93.0         | 13.2                       |

**Table 8.** Comparative performance of deep learning models for AQI prediction (Dataset 1).

Conclusion

An effectual air quality prediction with a health assessment system was designed to predict the pollutant level in the air in the surrounding environment. The developed method was also used for predicting the AQI. Employing the IGOA for feature selection and classification, this framework effectively identifies the most pertinent features from unprocessed air quality data, assigns appropriate weights, and utilizes these findings to enhance the predictive accuracy of the ARBi-LSTM-PD network. The approach prioritizes the maximization of variance and correlation coefficients during feature selection, ensuring that only the most significant features were included, which leads to a refined input for classification processes. Additionally, the adaptive structure of the network, augmented with pyramid dilation techniques, further enhances the model’s learning ability and predictive effectiveness. Through comprehensive optimization strategies proposed framework achieves notable enhancements in accuracy, NPV, and FNR. Using Dataset 1, the proposed model attained an accuracy of 95.1% with a sigmoid activation function. These findings highlight the potential of this cohesive methodology to deliver timely and trustworthy air quality evaluations, which are essential for public health initiatives. Thus, the advanced predictive capability of this method provides more precise forecasting of air quality. This gives policymakers the ability to take proactive measures to mitigate air pollution impacts before they reach critical levels. Timely and accurate air quality predictions help health officials to provide targeted health advisories. By reducing pollution levels and preventing harmful health outcomes through more accurate forecasting and the economic costs due to air pollution also reduced.

Practical implications

The developed air quality prediction model is used in urban planning and environmental policy and it is highly suitable in environmental mitigation efforts and assessing health risks in urban areas. While the primary focus of this study is on improving the predictive performance of air quality assessment, the scalability and real-time applicability of the proposed model are critical for deployment in smart environments. The IGOA-RBi-LSTM-PD framework is designed as a two-stage system, with computationally intensive components (such as weighted feature selection via IGOA and parameter tuning) executed offline during the training phase. The trained ARBi-LSTM-PD model then functions as a lightweight inference engine, enabling real-time predictions. To assess latency performance, the trained model was evaluated on a system equipped with an Intel Core i7 (11th Gen), 16GB RAM, and NVIDIA RTX 3060 GPU. The inference time for a single AQI instance was approximately 13.2 milliseconds, while batch inference (128 samples) was achieved in 174 milliseconds. These timings confirm that the model is suitable for real-time air quality monitoring applications, where inference windows typically range from 1 to 5 seconds. Additionally, the consistent performance across three distinct datasets confirms the model’s ability to generalize across sensor variations and data resolutions. This indicates high horizontal scalability to diverse urban regions or sensor infrastructures. For real-world deployments, further model compression techniques such as layer pruning, quantization, and deployment on edge-AI hardware (e.g., NVIDIA Jetson Nano, Coral Edge TPU) can be explored to reduce computational overhead and power consumption. Moreover, inference engines like ONNX or TensorRT can facilitate low-latency serving, further supporting real-time applications.

To support real-world applicability, the proposed model can be deployed using a hybrid edge-cloud architecture. The IGOA-driven optimization and model training are performed offline in the cloud, while the lightweight ARBi-LSTM-PD inference module can be deployed on edge devices using frameworks like TensorRT or ONNX for real-time predictions. The model supports update frequencies ranging from 5 seconds to 1 minute, aligning with typical air quality monitoring cycles. Moreover, it is compatible with city-wide IoT sensor networks, allowing integration with environmental data hubs to provide dynamic air quality assessments and health advisories across urban regions.

Research limitations and future scopes

The features and their corresponding weights are restricted to defined intervals, which reduces the model’s adaptability and decreases the important features or weight values that fall outside these specified boundaries. The effectiveness of the model will be influenced by the quality and representativeness of the input data. If the data contains biases or inaccuracies, it could result in less effective feature selection and classification results. Incorporating pyramid dilation within the ARBi-LSTM might add extra computational complexity, potentially hindering the model’s scalability and efficiency, especially when working with larger datasets. To overcome these difficulties future scopes will generate adaptive methods for the ranges of features and weights that will modify based on initial findings, which enhance the overall flexibility and performance of the optimization approach. Additionally, applying multi-objective optimization strategies to optimize multiple goals simultaneously will lead to a more balanced prediction model.

## Major findings of this proposed model

The combination of optimized weighted feature selection and sophisticated classification methods greatly improves the precision of air quality forecasts while also increasing the significance of health evaluations. This advancement plays a crucial role in fostering more efficient environmental monitoring and public health initiatives. By connecting air quality forecasts with health outcomes, proposed framework offers a systematic mechanism for evaluating the effects of environmental factors on public health.

## Data availability

<https://github.com/adityarc19/aqi-india>; <https://data.gov.in/resource/real-time-air-quality-index-various-locations>; <https://github.com/Anindya-Das02/Comparison-of-ML-models-for-predicting-AQI>.

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## Author contributions

R Sudha – Research proposal – construction of the work flow and model – Final Drafting; Ajith Damodaran – Survey of Existing works – Improvisation of the proposed model; Gunaselvi Manohar– Initial Drafting of the paper – Collection of datasets and choice of their suitability – Formulation of pseudocode.

## Declaration

## Competing interests

The authors declare no competing interests.

## Additional information

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